VIRTUATILT - VISUAL PINBALL X SETUP

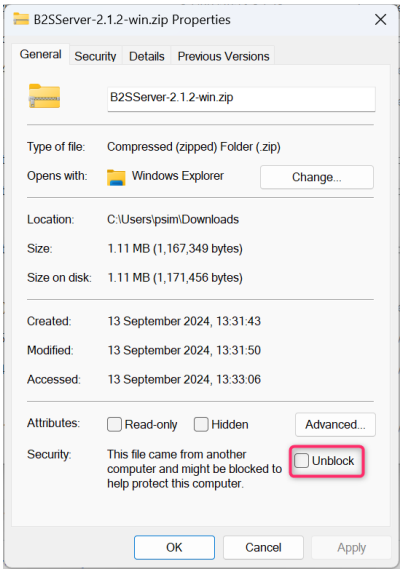
Following some basic instructions to have your new VirtuaTilt up and running with Visual Pinball X.

DOF Installation

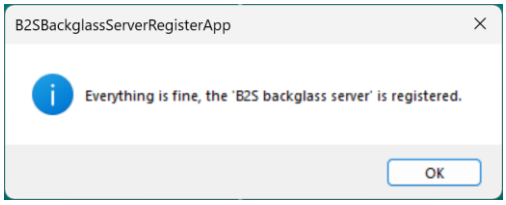
IMPORTANT: DOF (Direct Output Framework) is needed if you want to experience a real pinball simulation with all feedback devices in Visual Pinball X. Latest DOF version is required for Pinscape Pico support: http://mjrnet.org/pinscape/dll-updates.html

You need **B2S Server** installed first. If you have already installed it, skip this part.

For first-time users, download latest version from here: **https://github.com/vpinball/b2s-backglass/releases** and unblock it:

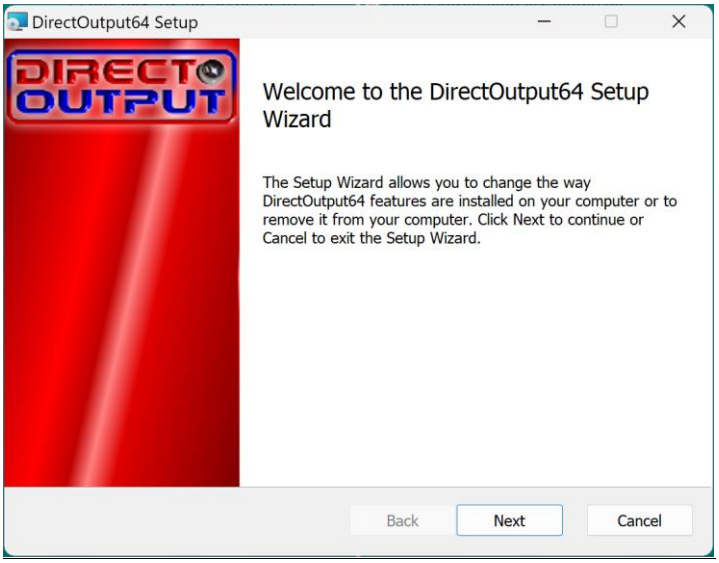


Unzip the B2SServer archive in VisualPinball folder (typically **C:\vPinball\VisualPinball\Tables**) and run the **B2SBackglassServerRegisterApp.exe** as Administrator (right-click). This registers the software as a COM object:

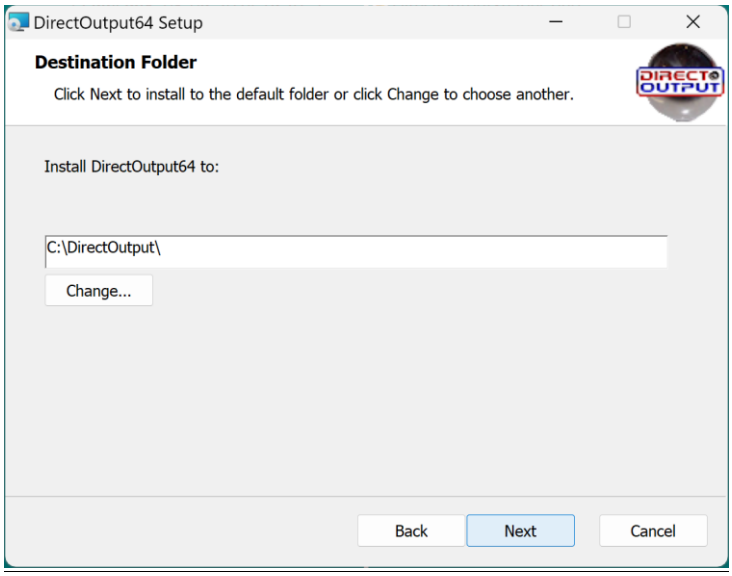


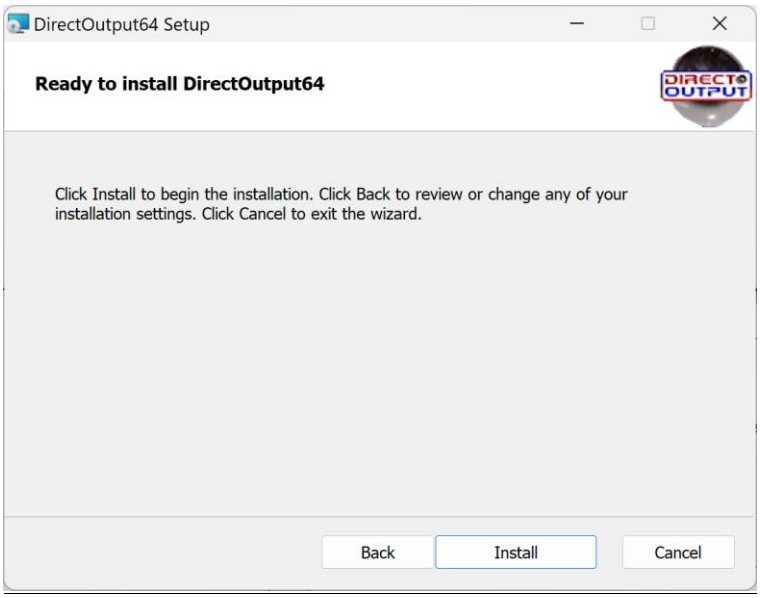
Latest version of DOF allows merging of X86 and X64 installs, so you can safely install both versions.

Install DOF using .msi files from official page: **http://mjrnet.org/pinscape/dll-updates.html** and launch it:

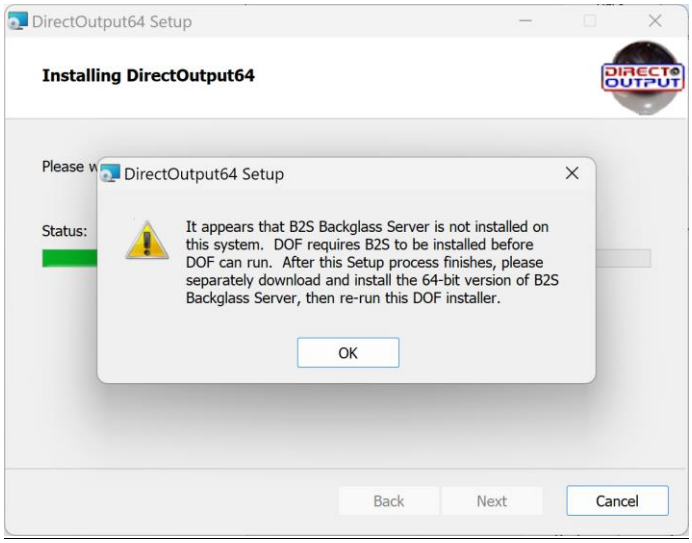


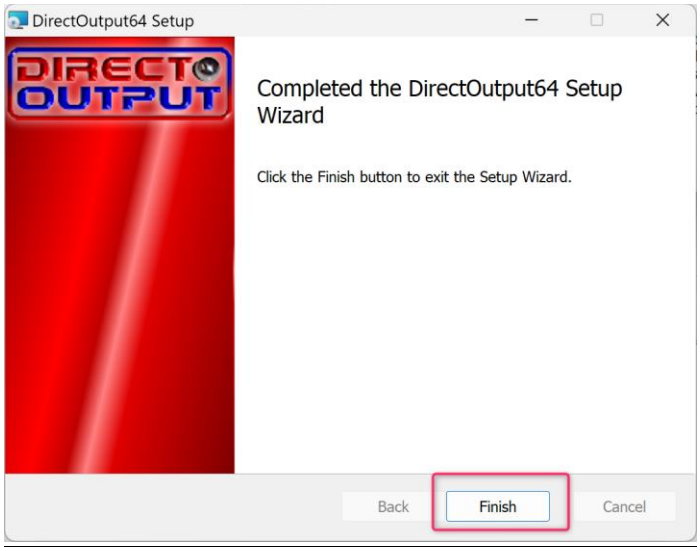
Leave installation folder as-is (**C:\DirectOutput**). Highly recommended:



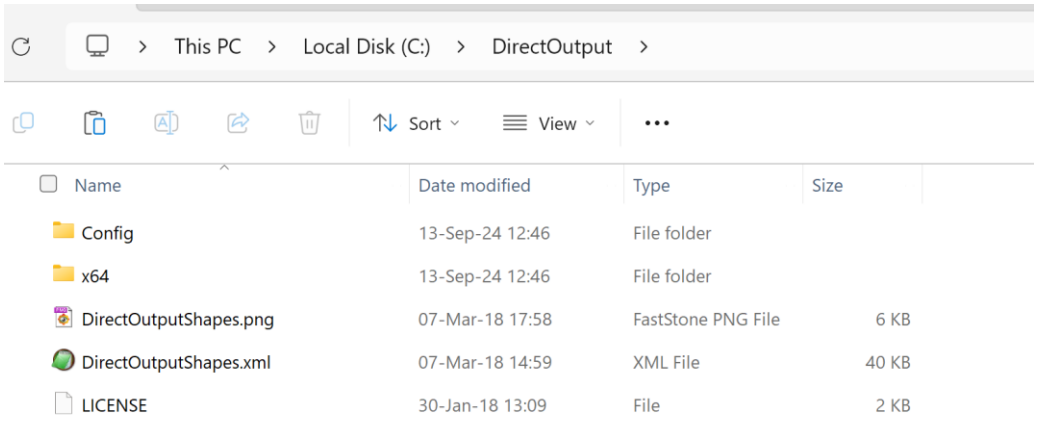


Do not care about the following error (it should appear if you don't have B2S Server installed) and click OK:

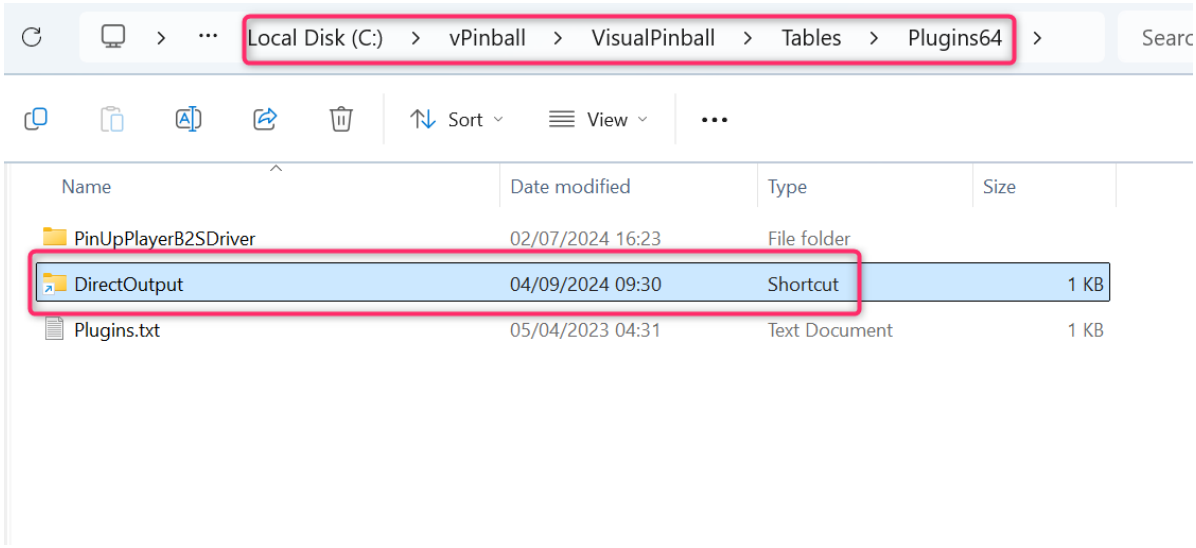




This is how the **\DirectOutput** folder will look like:



A shortcut to DirectOutput folder in **\Plugins64** folder of B2SServer installation should be present (created by DOF setup):

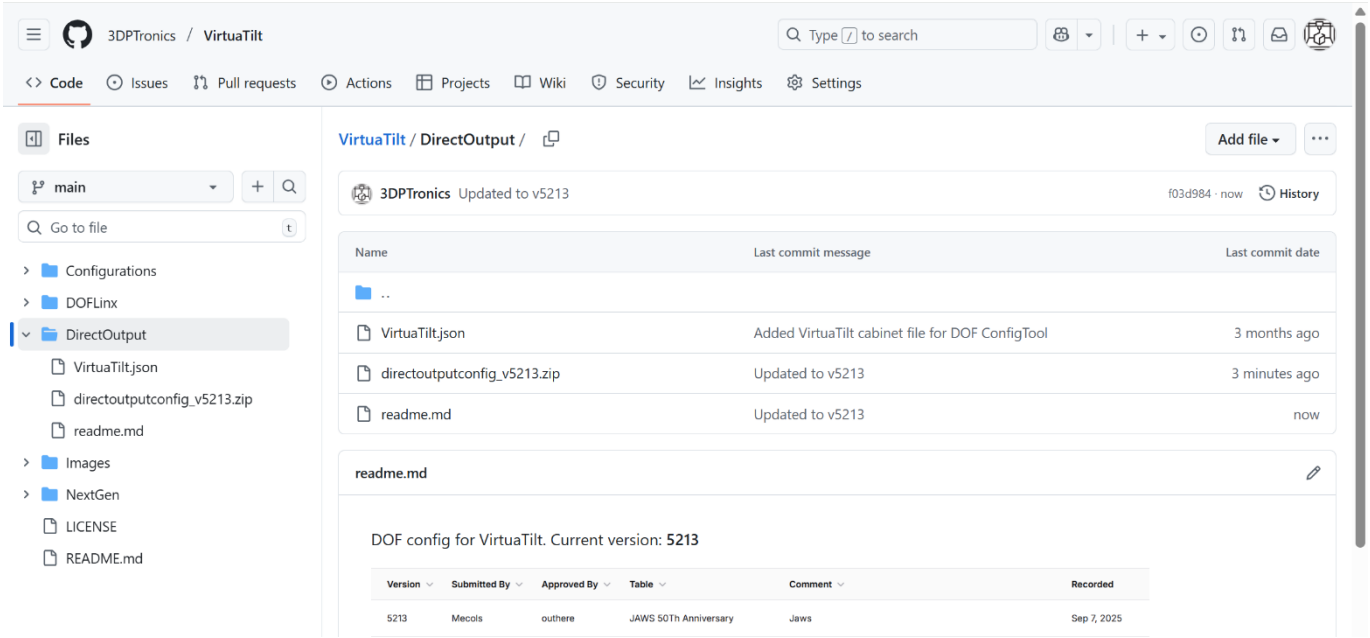


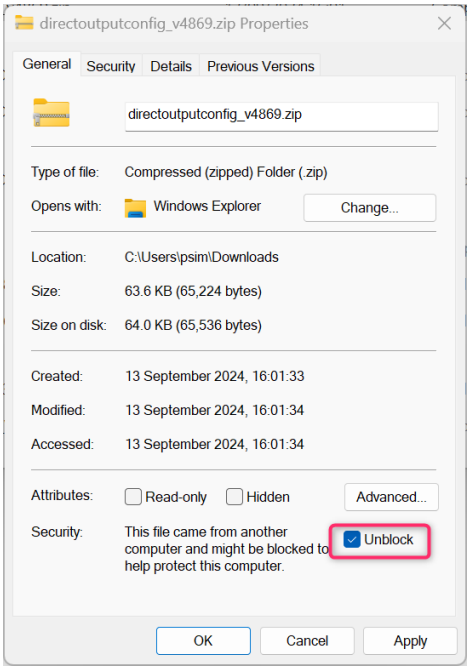
If not, manually create a shortcut to **DirectOutput** folder and put it in **Plugins64** folder of B2SServer installation.

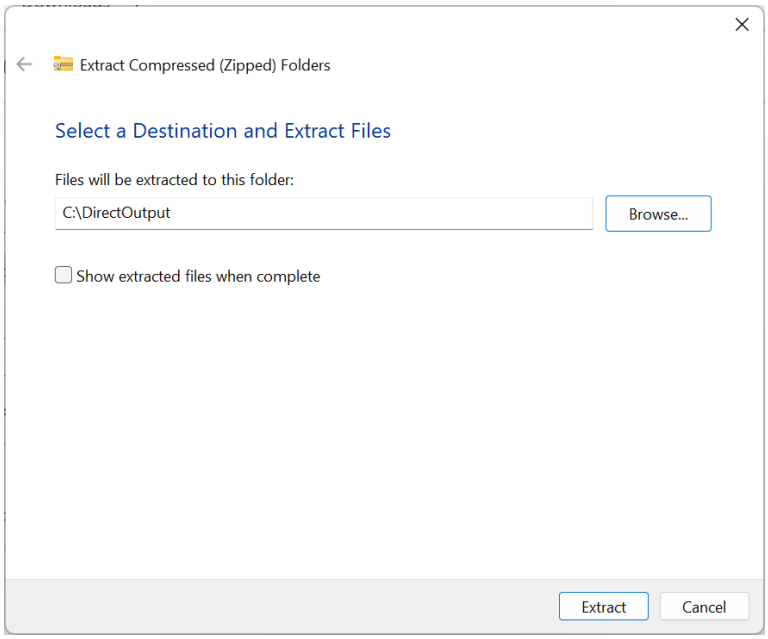
As a reference, the official step-by-step guide for DOF configuration is available here: Pinscape Build Guide (mjrnet.org).

Configure DOF for VirtuaTilt

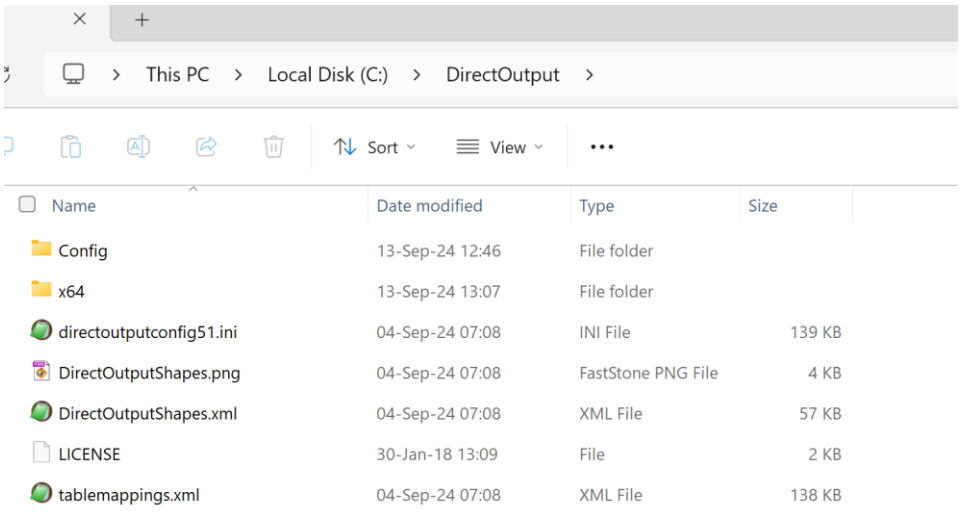
To configure DOF you need to set up a configuration file. You can create an account and login to the DOF Config tool webpage or **you can just download provided config files from our GitHub repository (including cabinet file)**, unblock the file and copy/move the files into **C:\DirectOutput** directory (or wherever DOF is installed on your PC), overwriting existing files:



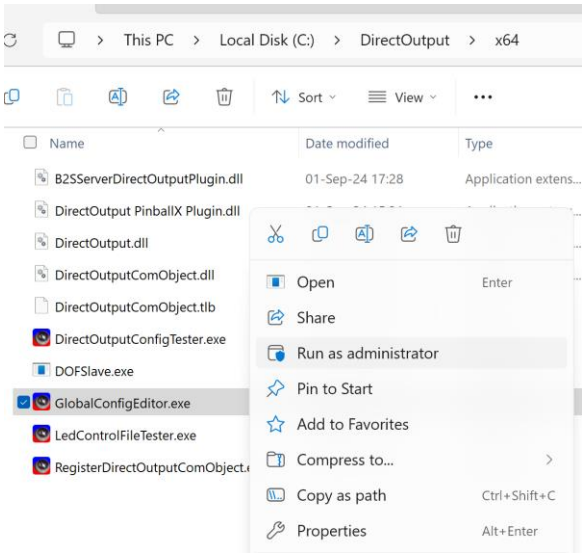




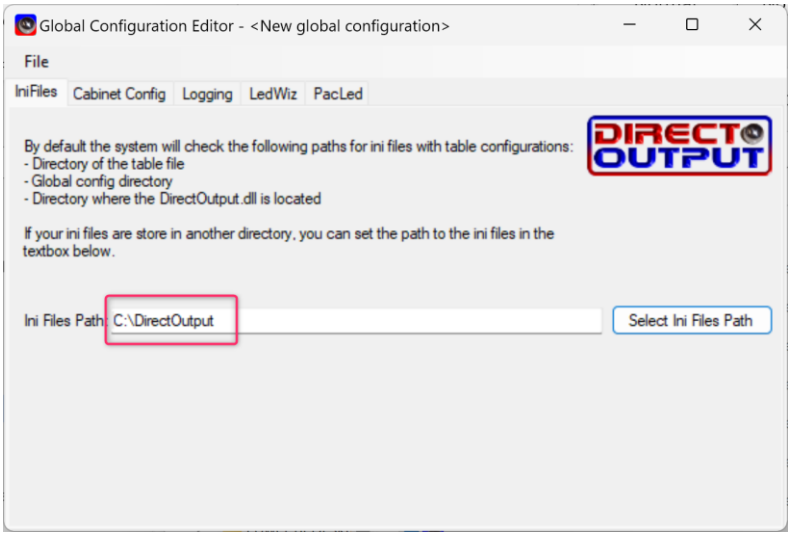
This is how the **DirectOutput** folder will look like:



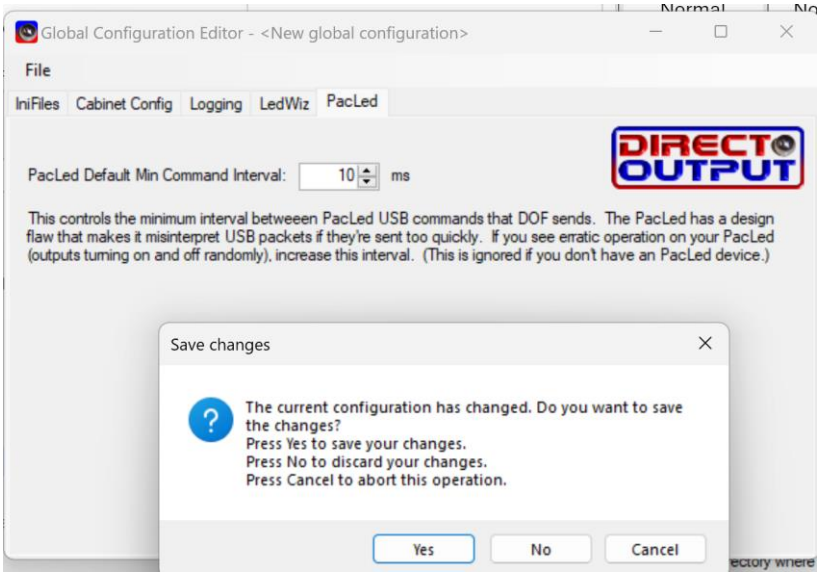
Now right-click **GlobalConfigEditor.exe** file in **/DirectOutput/X64** and in **/DirectOutput/X86** folder and launch it as Administrator:



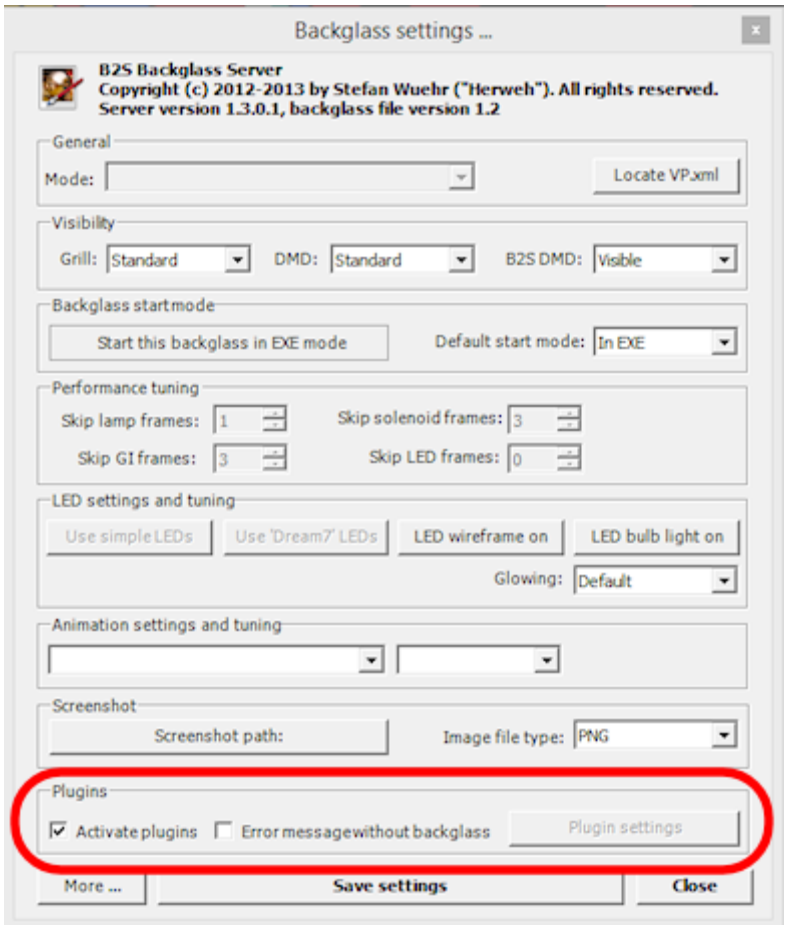
Configure .ini file path where you previously unzipped our config file (**C:\DirectOutput**):



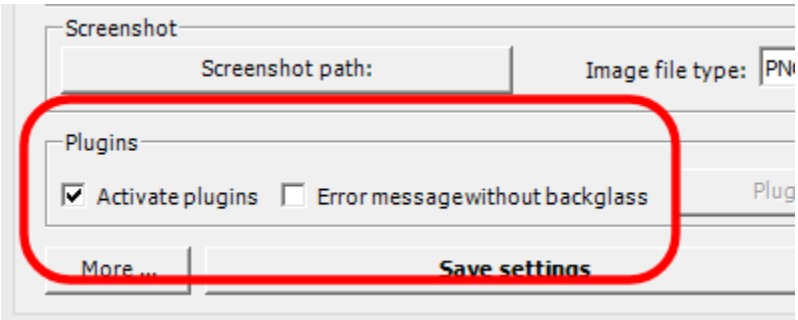
Close the window and press **Yes** to save all changes:



Start Visual Pinball and load a table that has a matching B2S backglass file installed; right-click the mouse anywhere in the backglass area to bring up the B2S options dialog:



Check the box **Activate plugins** and uncheck the box **Error message without backglass**.

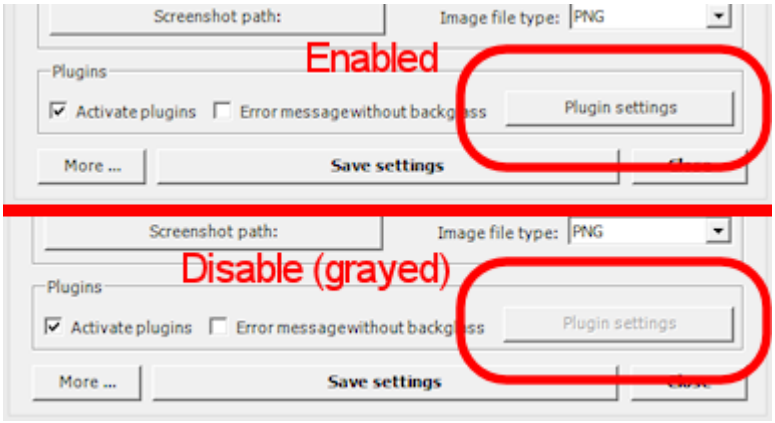


*Make sure that **Activate plugins** flag is checked and **Error message without backglass** flag is un-checked*

- Click Save Settings
- Exit the table (press "Q" and then "Q" again) and close VP

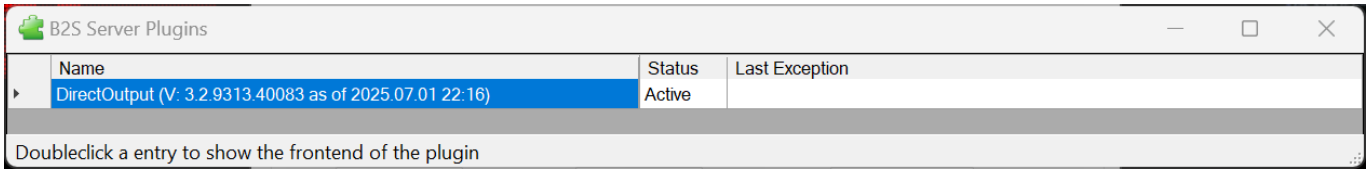
Now start VisualPinball and load a table that has a B2S backglass. You can use the same table you used during the setup procedure in the step where we updated the B2S backglass settings.

As before, when the backglass appears, right-click the mouse anywhere in the backglass display area to bring up the options dialog. Look to see if the **Plugins** button at the bottom is **enabled**:



If the button is disabled, DOF isn't getting loaded. Go to the official troubleshooting section for things to try.

If the button is enabled, click it. This will bring up a separate dialog that shows the status of each plugin.



Look for a **DirectOutput** entry in the list. If you don't see any such entry, it means the same thing as a disabled Plugin Status button, namely that DOF isn't being loaded. Go to the official troubleshooting section for help.

Finally, check the **Status** and **Last Exception** columns for the Direct Output entry.

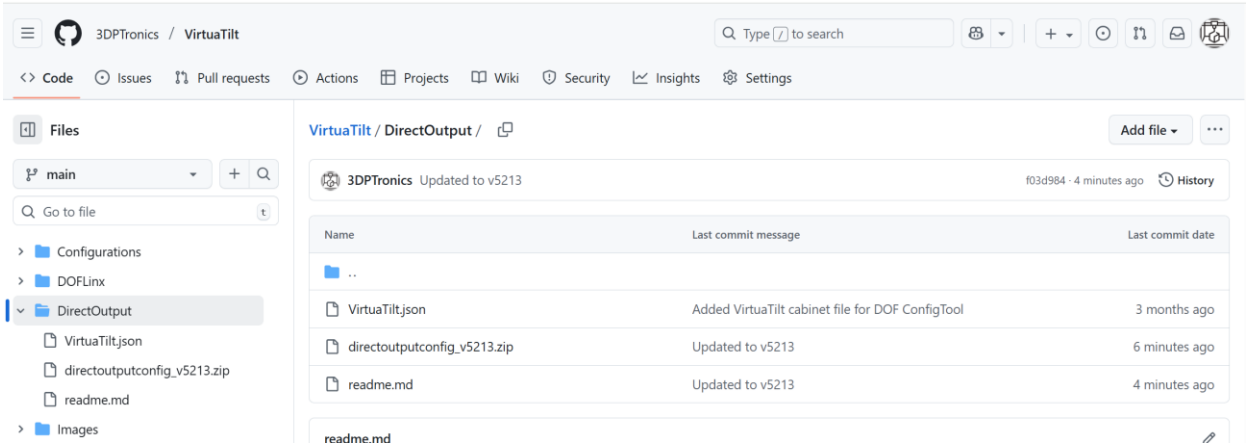
If the Status is **Disabled**, or there's a message in the Last Exception box, see the official troubleshooting section for help.

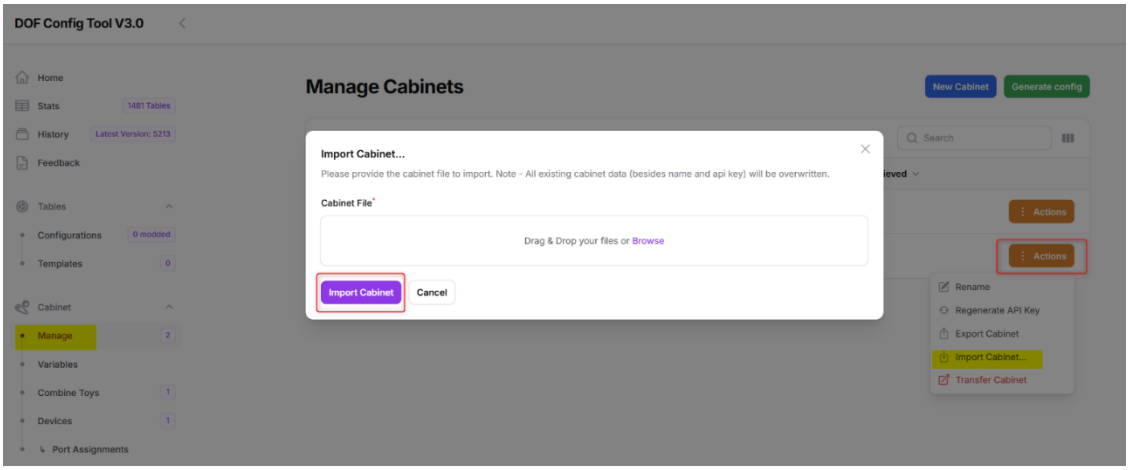
If the Status is **Active** and the Last Exception column is empty, congratulations! Your DOF setup work was successful! DOF is loading and starting correctly.

DOF Configuration Change

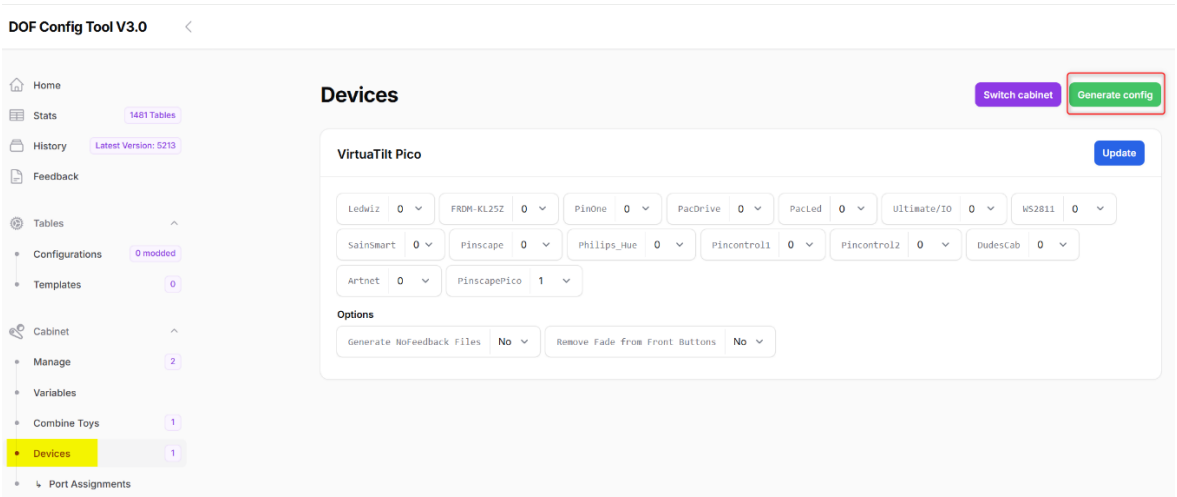
IMPORTANT NOTE: If you don't want to mess with following DOF configuration, you can just download **VirtuaTilt cabinet file** from our GitHub and import it on DOF Config Tool:

https://github.com/3DPTronics/VirtuaTilt/tree/main/DirectOutput





Once imported, you can generate your updated config file which contains all latest tables:

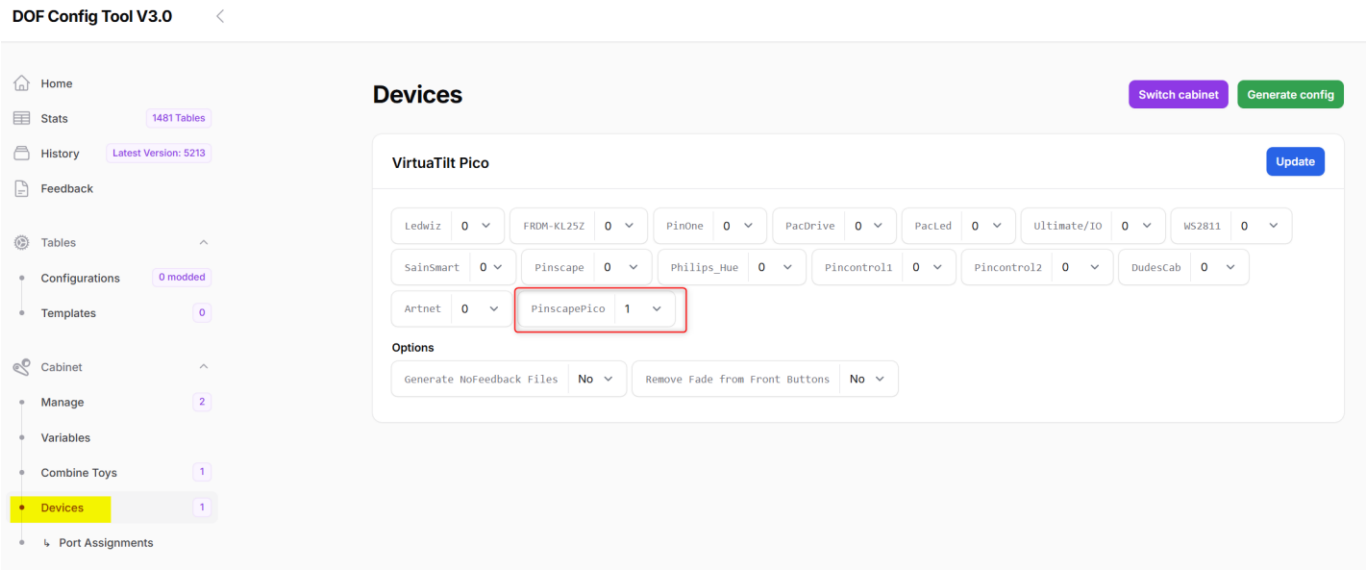


Select your device

If you want to change the default DOF configuration for VirtuaTilt, you need to create an account and login to the DOF Config tool page.

Once you have an account setup, go to **Devices** and select the number of devices for your setup.

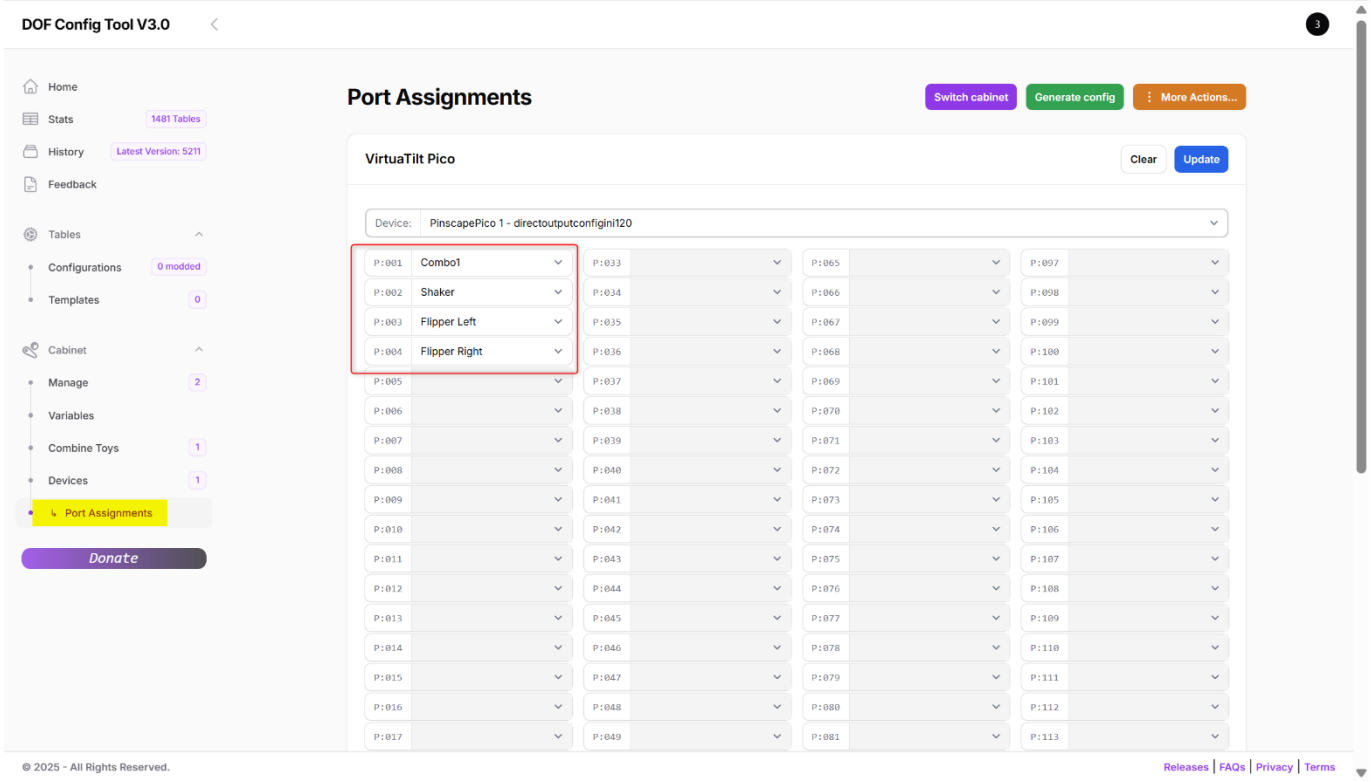
- Select **1** for number of **PinscapePico** devices and then click on **Update:**



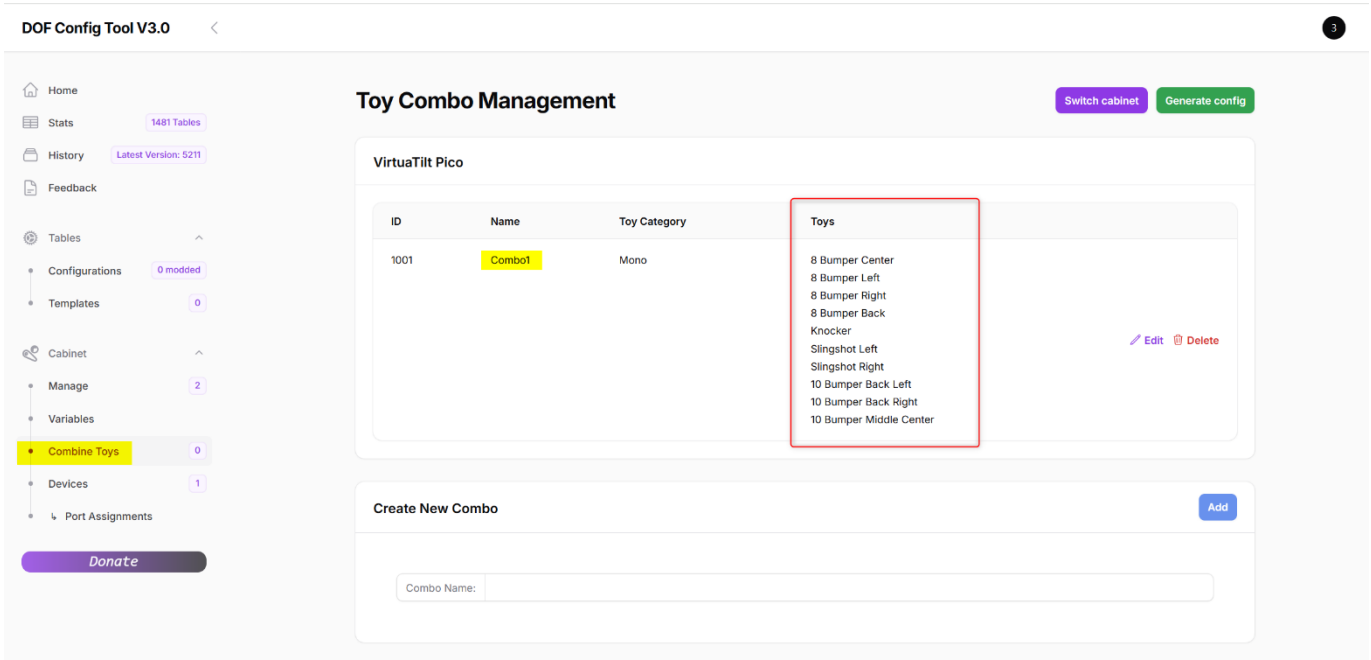
Creating a config file

When creating a config file for your setup, it's simply a matter of selecting the outputs for each device in your cabinet. VirtuaTilt is already configured this way:

- **Port1** – Solenoid – Combo for bumpers and slingshots
- **Port2** – Shaker Motor
- **Port3** – Left Flipper Solenoid
- **Port4** – Right Flipper Solenoid

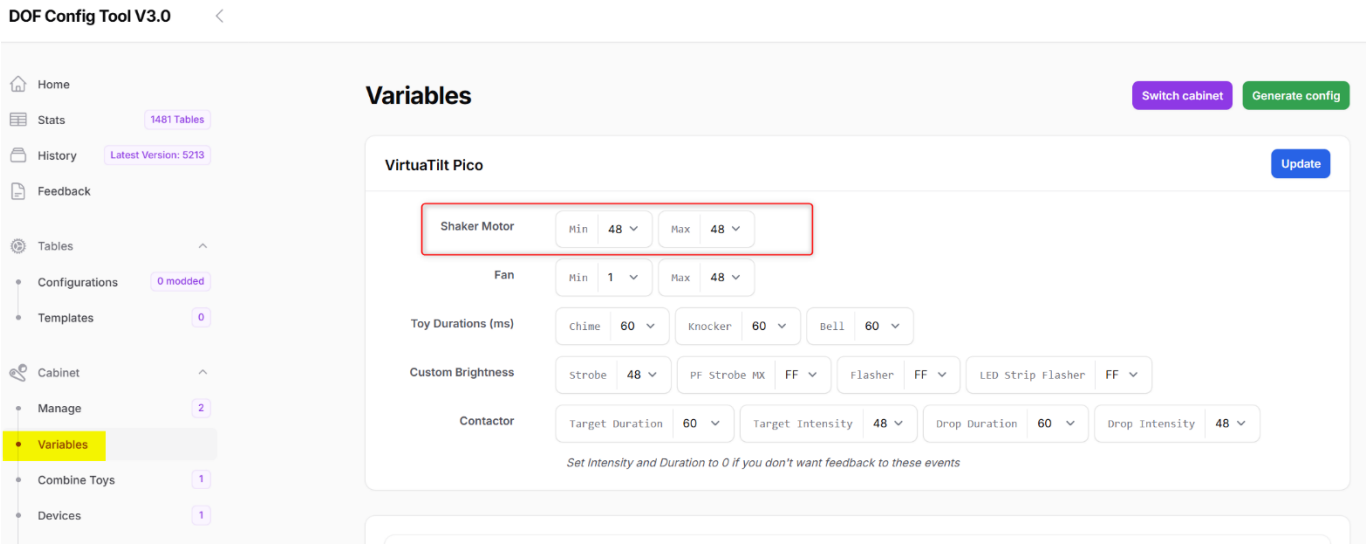


The **Combo1** config is assigned to knocker, slingshots and bumpers:



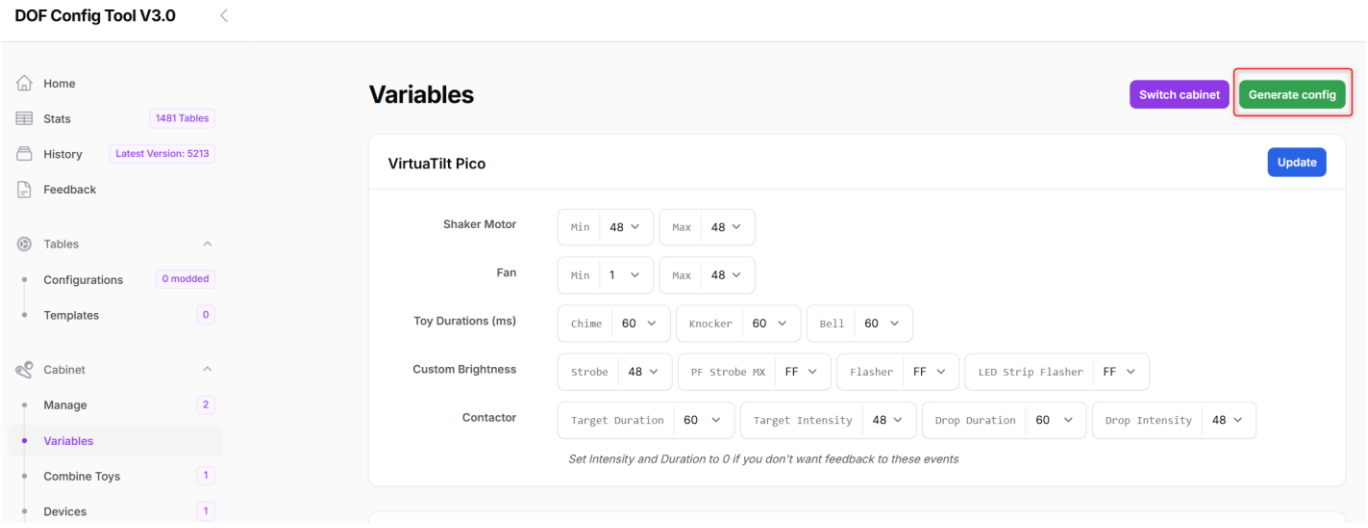
You can change it according to your personal preferences. To do this, simply navigate to the "**Combine Toys**" section in the config tool and enter your new combos.

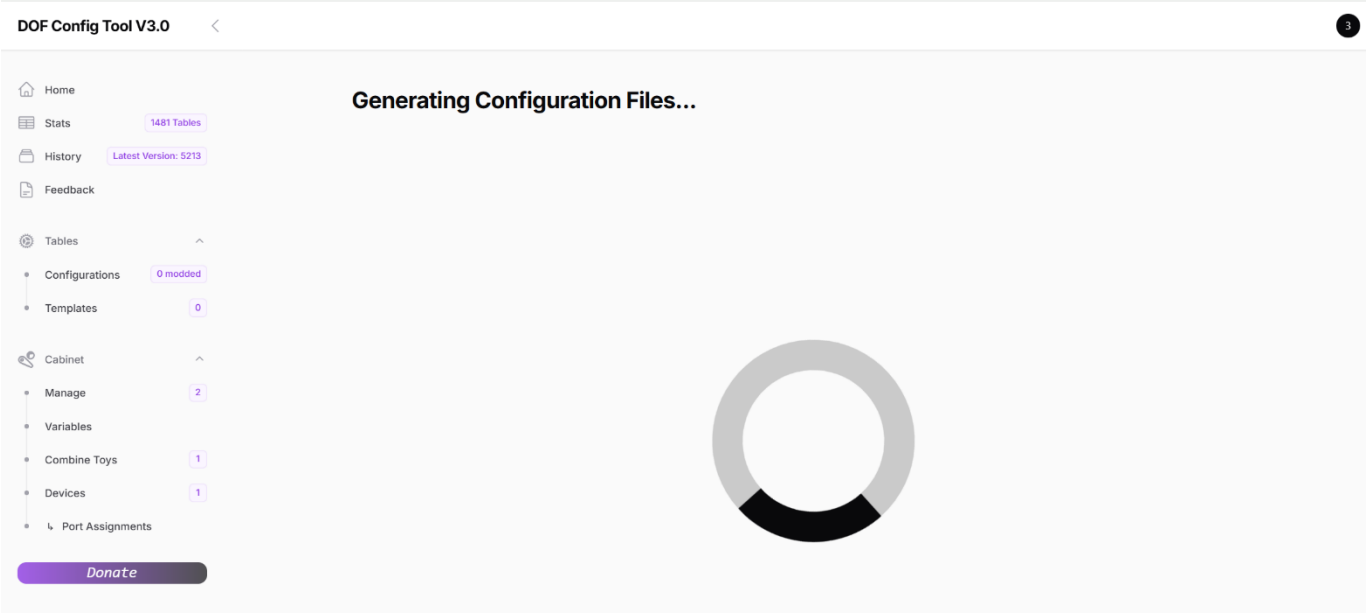
NOTE: Shaker Motor Intensity is already set at Maximum (**48**). You can change it according to your personal preferences:



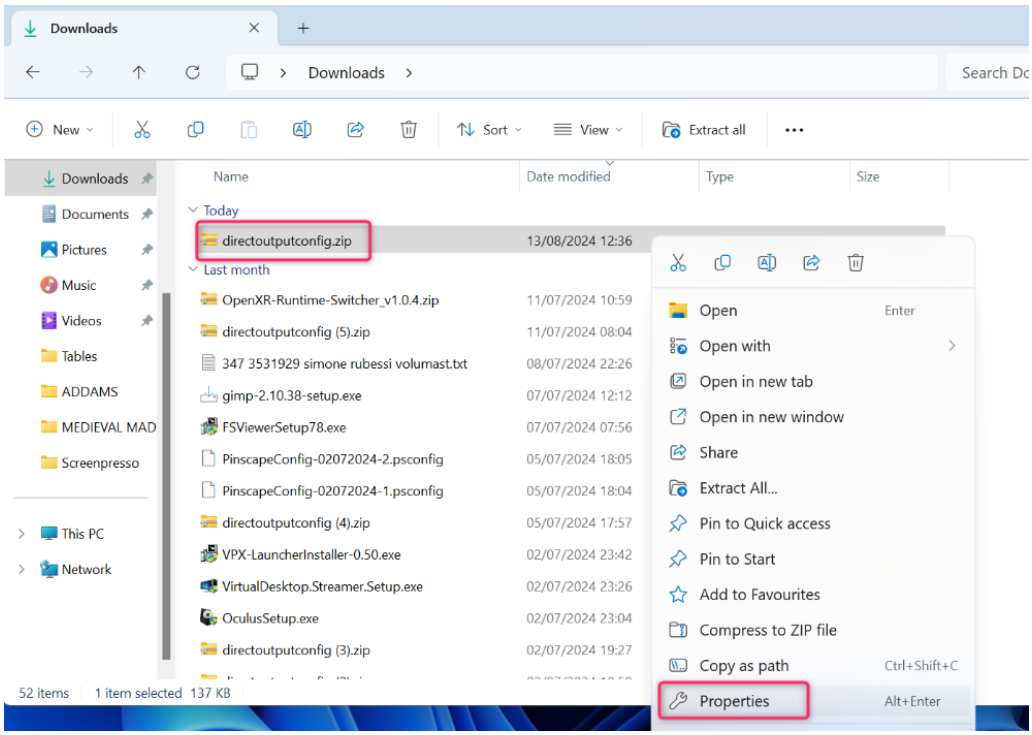
Generate and Import your DOF config

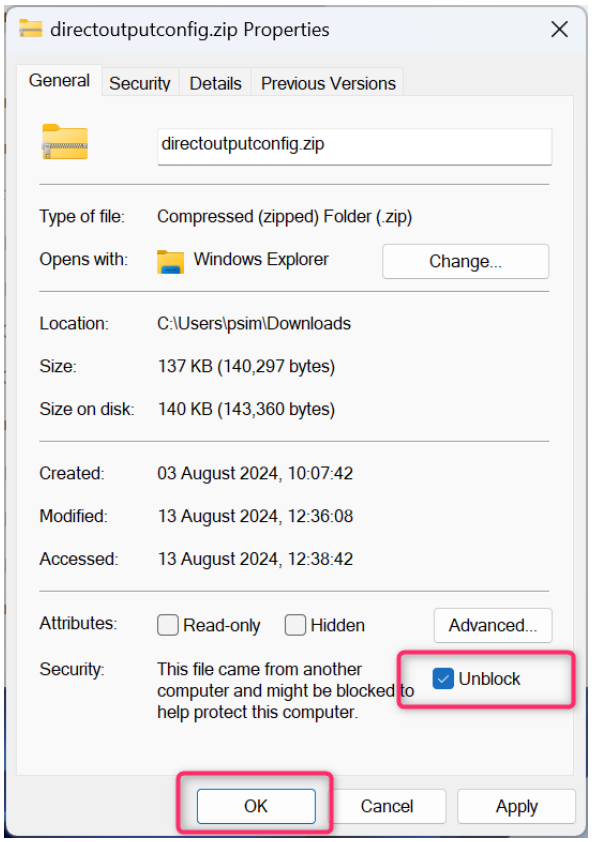
Once you have everything set in the config tool, click the **Update** button, then **Generate Config** button:

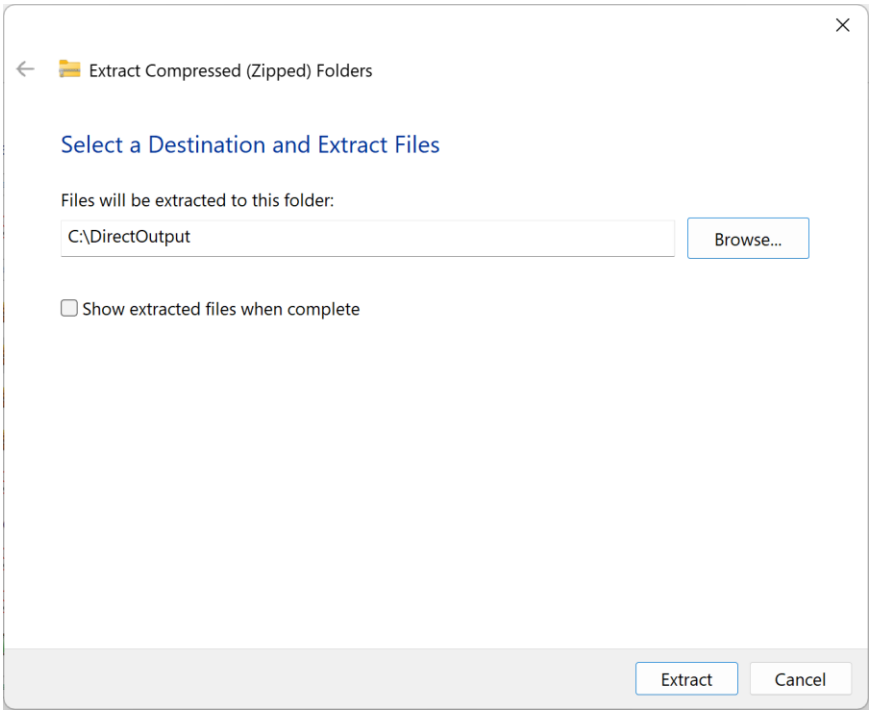


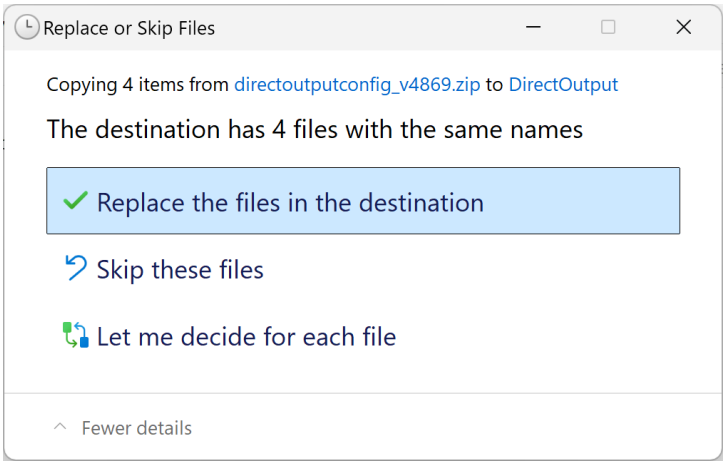


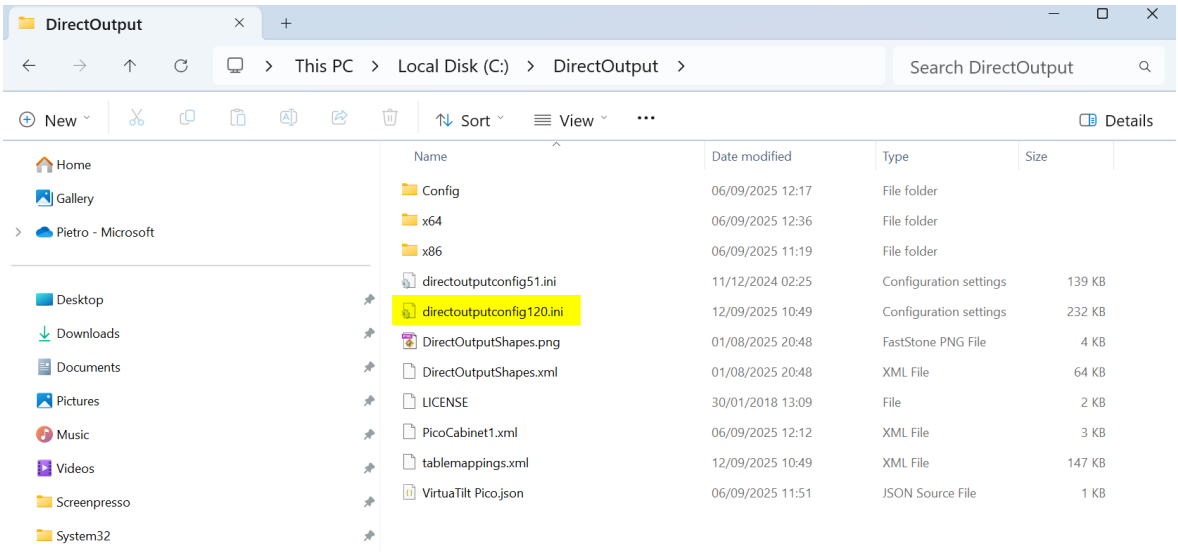
This will download a copy of the configuration files onto your PC. All you need to do now is to extract the zipfile into your **C:\DirectOutput** directory (or wherever DOF is installed on your PC) and overwrite all files:




directoutputconfig.zip Properties
General
Security
Details
Previous Versions
directoutputconfig.zip
Type of file:
Compressed (zipped) Folder (.zip)
Opens with:
Windows Explorer
Change...
Location:
C:\Users\psim\Downloads
Size:
137 KB (140,297 bytes)
Size on disk:
140 KB (143,360 bytes)
Created:
03 August 2024, 10:07:42
Modified:
13 August 2024, 12:36:08
Accessed:
13 August 2024, 12:38:42
Attributes:
Read-only
Hidden
Advanced...
Security:
This file came from another computer and might be blocked to help protect this computer.
Unblock
OK
Cancel
Apply

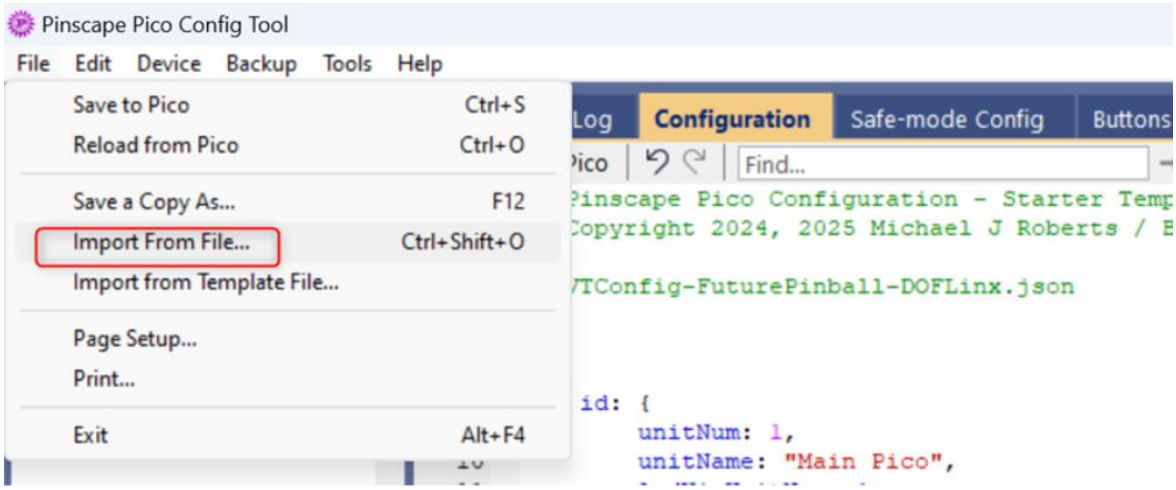

Extract Compressed (Zipped) Folders
Select a Destination and Extract Files
Files will be extracted to this folder:
C:\DirectOutput
Browse...
Show extracted files when complete
Extract
Cancel

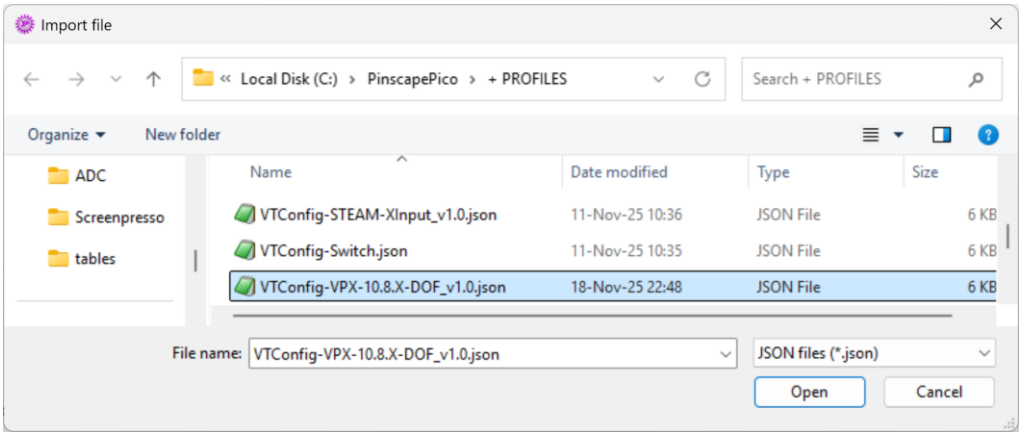

Replace or Skip Files
Copying 4 items from directoutputconfig_v4869.zip to DirectOutput
The destination has 4 files with the same names
Replace the files in the destination
Skip these files
Let me decide for each file
Fewer details


DirectOutput
This PC
Local Disk (C:)
DirectOutput
Search DirectOutput
New
Sort
View
Details
Home
Gallery
Pietro - Microsoft
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenpresso
System32
Name
Date modified
Type
Size
Config 06/09/2025 12:17 File folder
x64 06/09/2025 12:36 File folder
x86 06/09/2025 11:19 File folder
directoutputconfig51.ini 11/12/2024 02:25 Configuration settings 139 KB
directoutputconfig120.ini 12/09/2025 10:49 Configuration settings 232 KB
DirectOutputShapes.png 01/08/2025 20:48 FastStone PNG File 4 KB
DirectOutputShapes.xml 01/08/2025 20:48 XML File 64 KB
LICENSE 30/01/2018 13:09 File 2 KB
PicoCabinet1.xml 06/09/2025 12:12 XML File 3 KB
tablemappings.xml 12/09/2025 10:49 XML File 147 KB
VirtuaTilt Pico.json 06/09/2025 11:51 JSON Source File 1 KB

Visual Pinball X - General Configuration

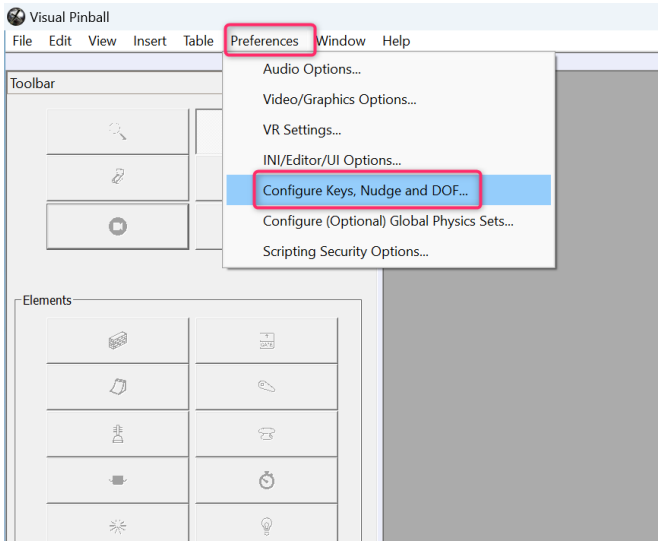
Load in Pinscape Pico Config Tool the **VTConfig-VPX-10.8.X-DOF_V1.X.json** profile first (**valid for all VPX versions**):



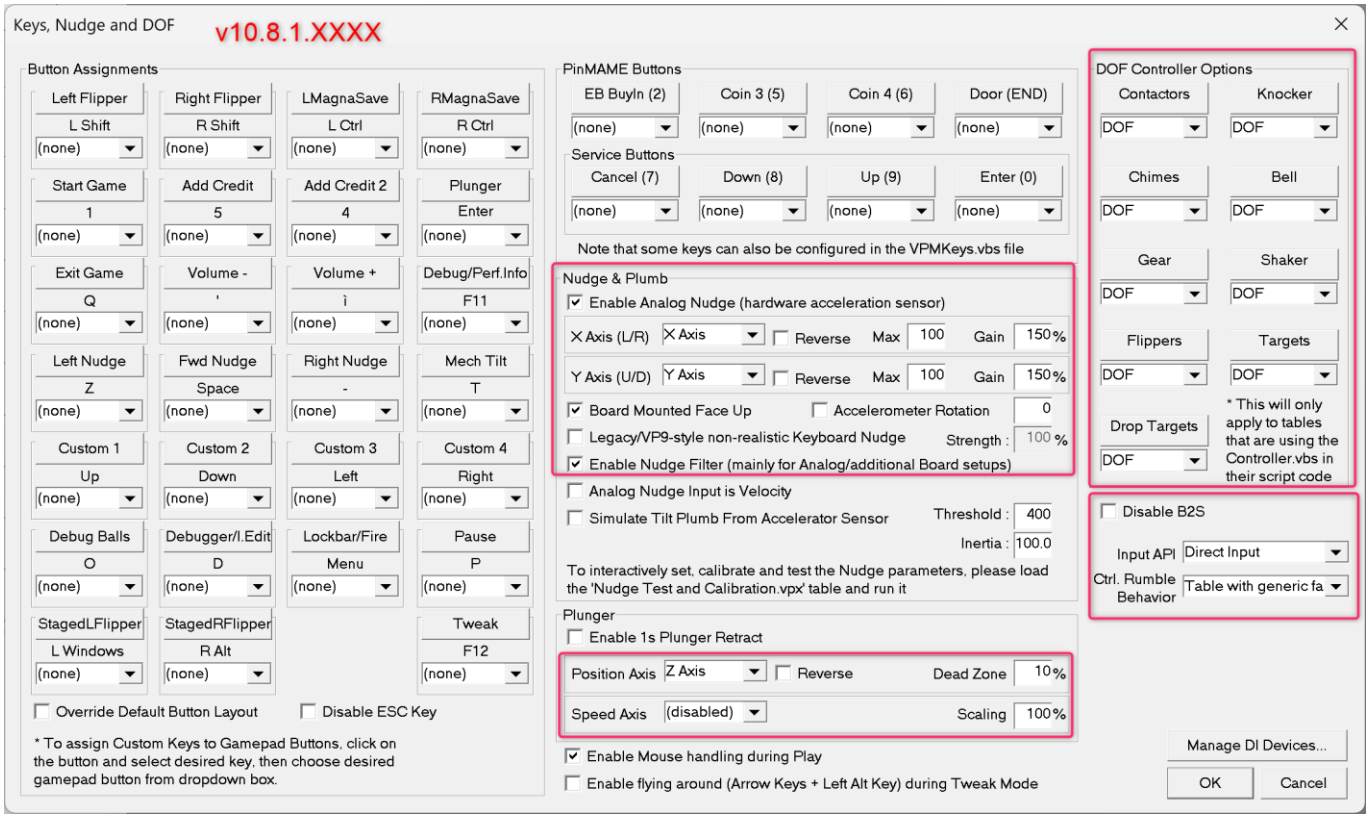




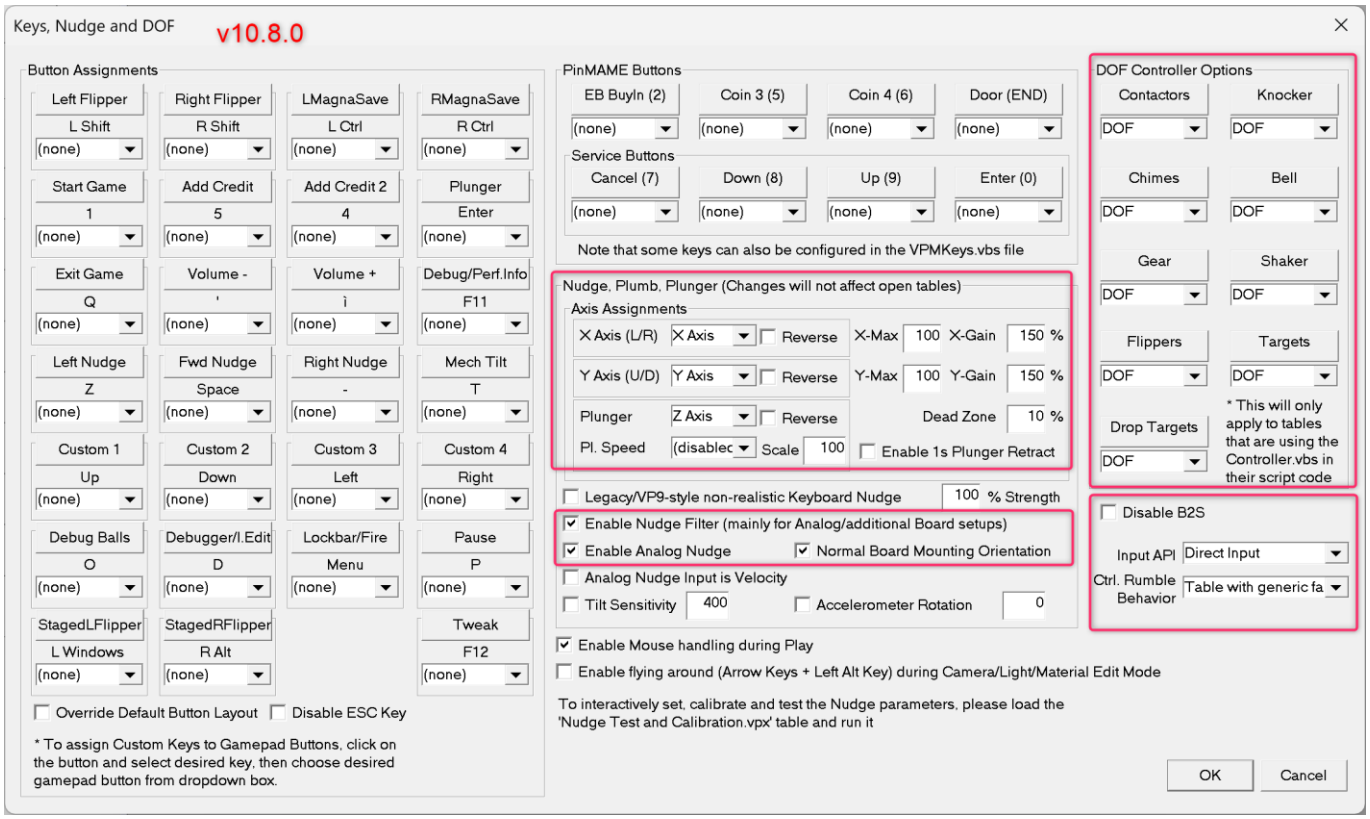
Replicate the following configuration in Visual Pinball X settings:



VPX 10.8.1.XXXX Settings



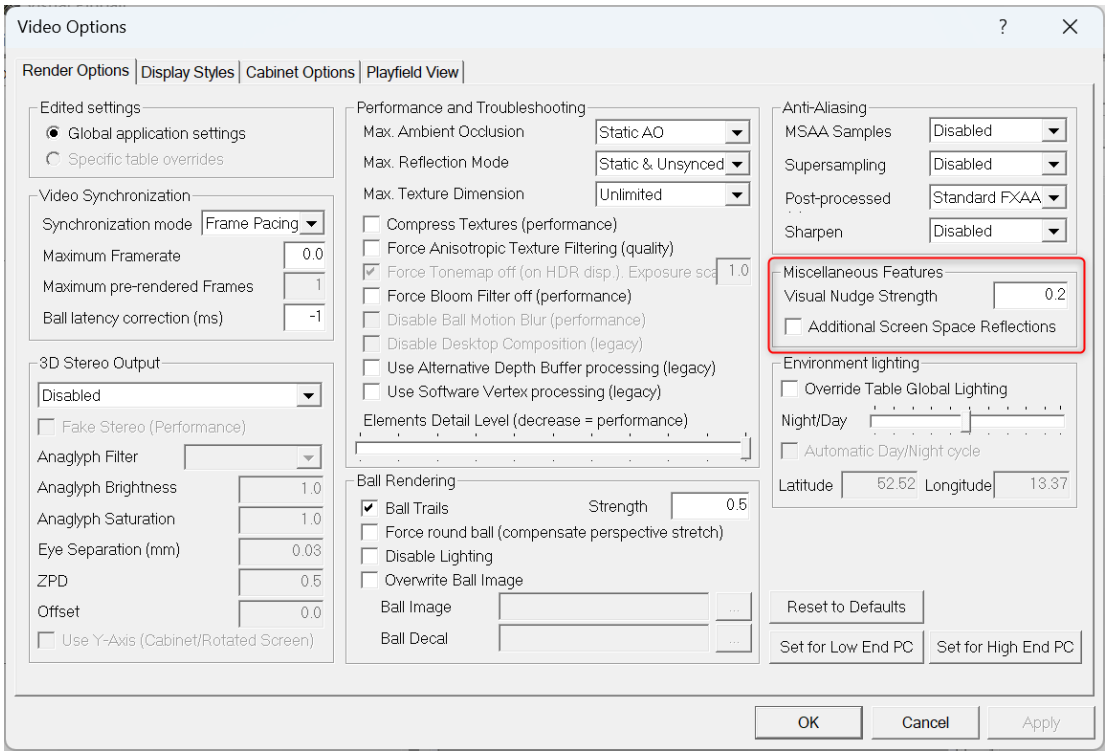
VPX 10.8.0 Settings



IMPORTANT NOTE: Depending on VPX versions (**https://github.com/vpinball/vpinball/releases**), you **may** need to select/unselect "**Analog Nudge Input Is Velocity**" to get nudge working. You **may** also need to switch **Input API** setting between **SDL / DirectInput.**

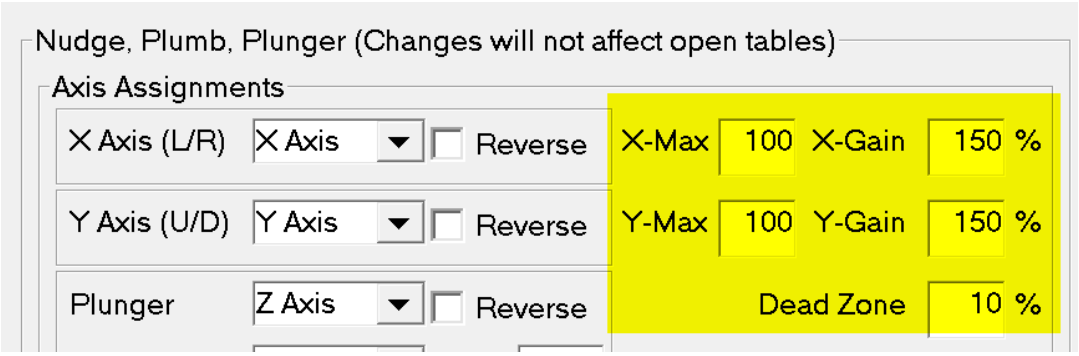
Visual Pinball X - Nudge/Tilt configuration

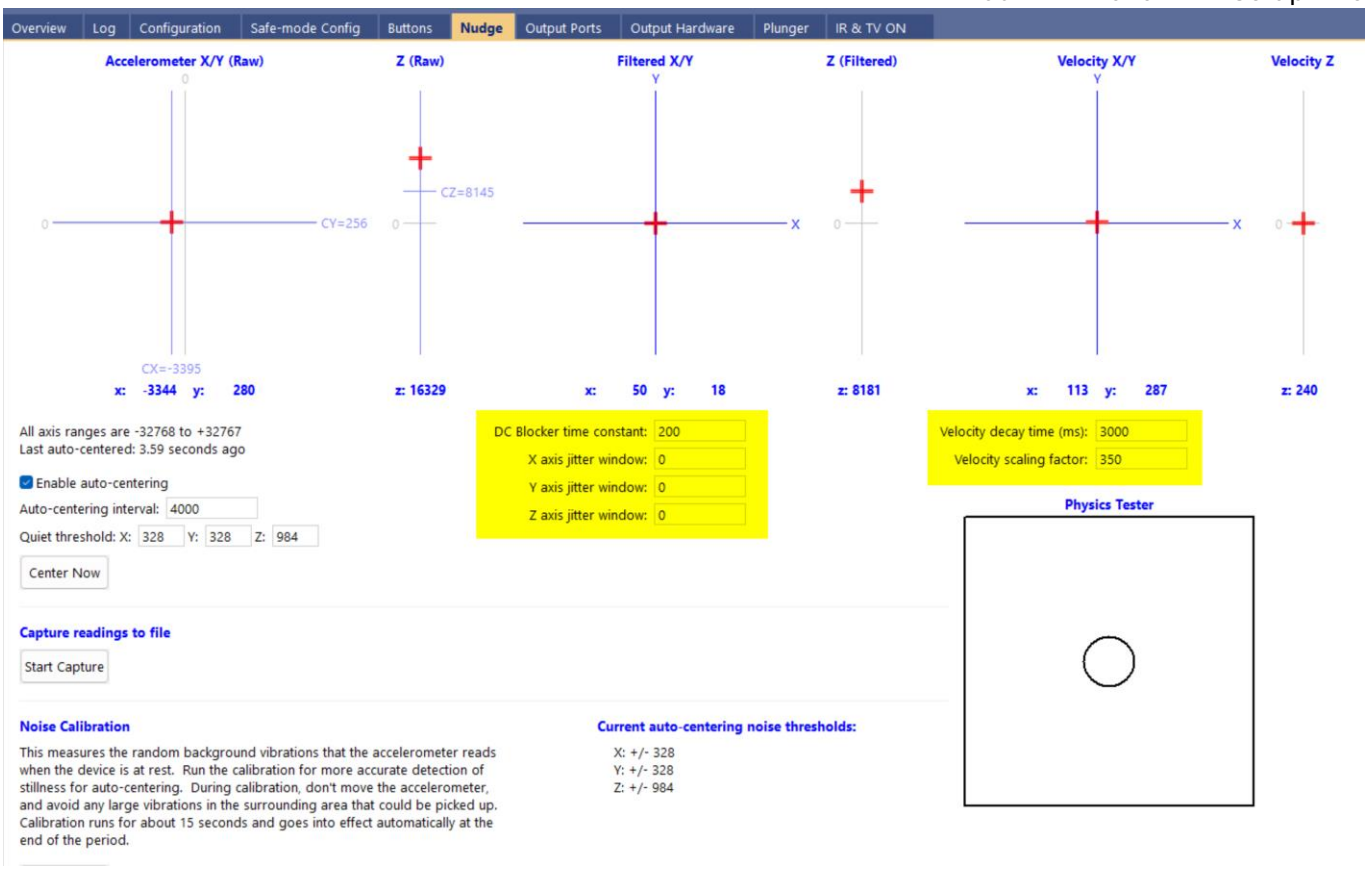
We recommend starting with the following values and modifying them afterwards:



Visual Nudge Strength: **0.2**

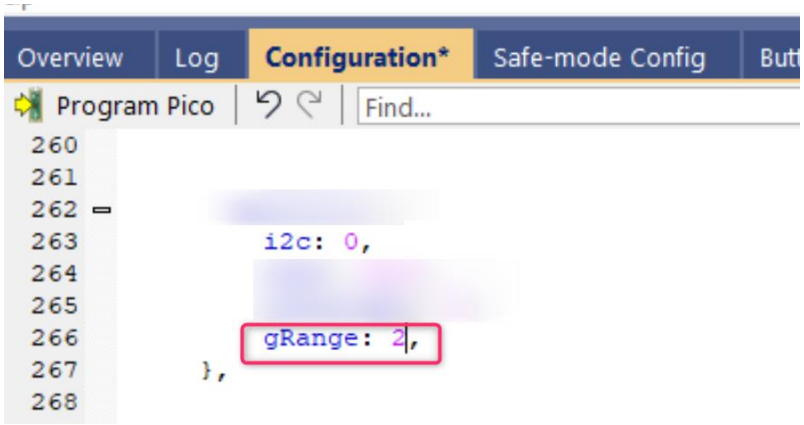
VERY IMPORTANT NOTE: You need to find your right settings (according to your personal preferences) for nudging. You can adjust both Pinscape Pico accelerometer settings and VPX settings until you find your perfect spot:





Try to adjust DC Blocker time constant, Velocity decay time and Velocity scaling factor.

You can also increase accelerometer sensitivity by changing gRange value to 2 in any profile:

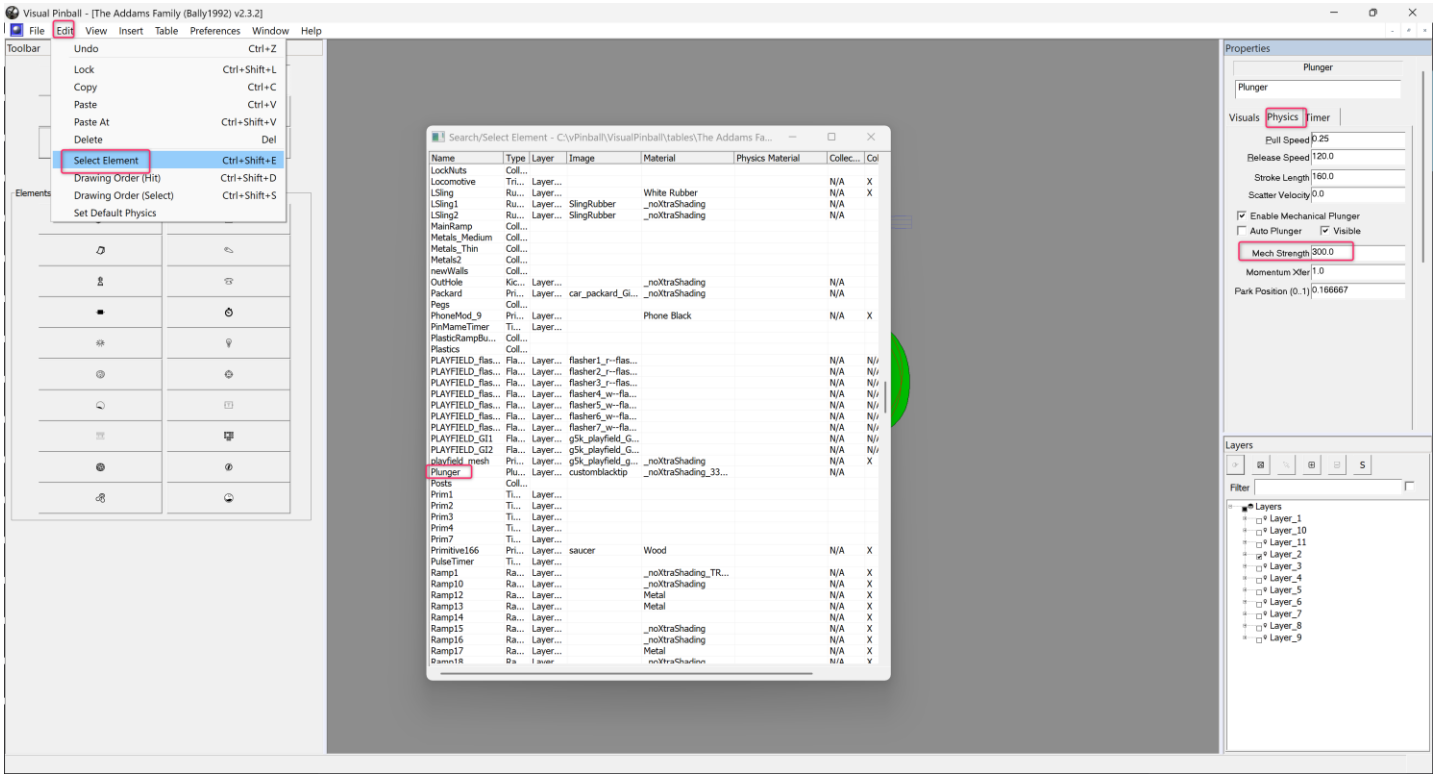


Please refer to the official Pinscape Pico reference for additional information about accelerometer parameters: http://mjrnet.org/pinscape/PinscapePico/Help/NudgeWin.htm

Visual Pinball X - Plunger configuration

If you feel that plunger is not strong enough for some VP tables, you can raise its **mechanical strength** in any table via software:

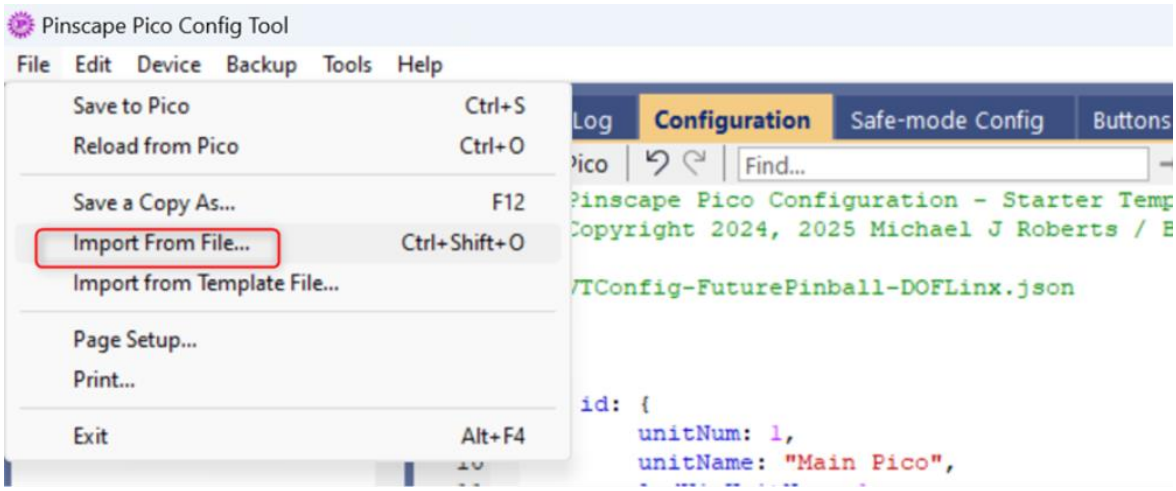
Edit → Select Element → Plunger → Physics → Mech Strength

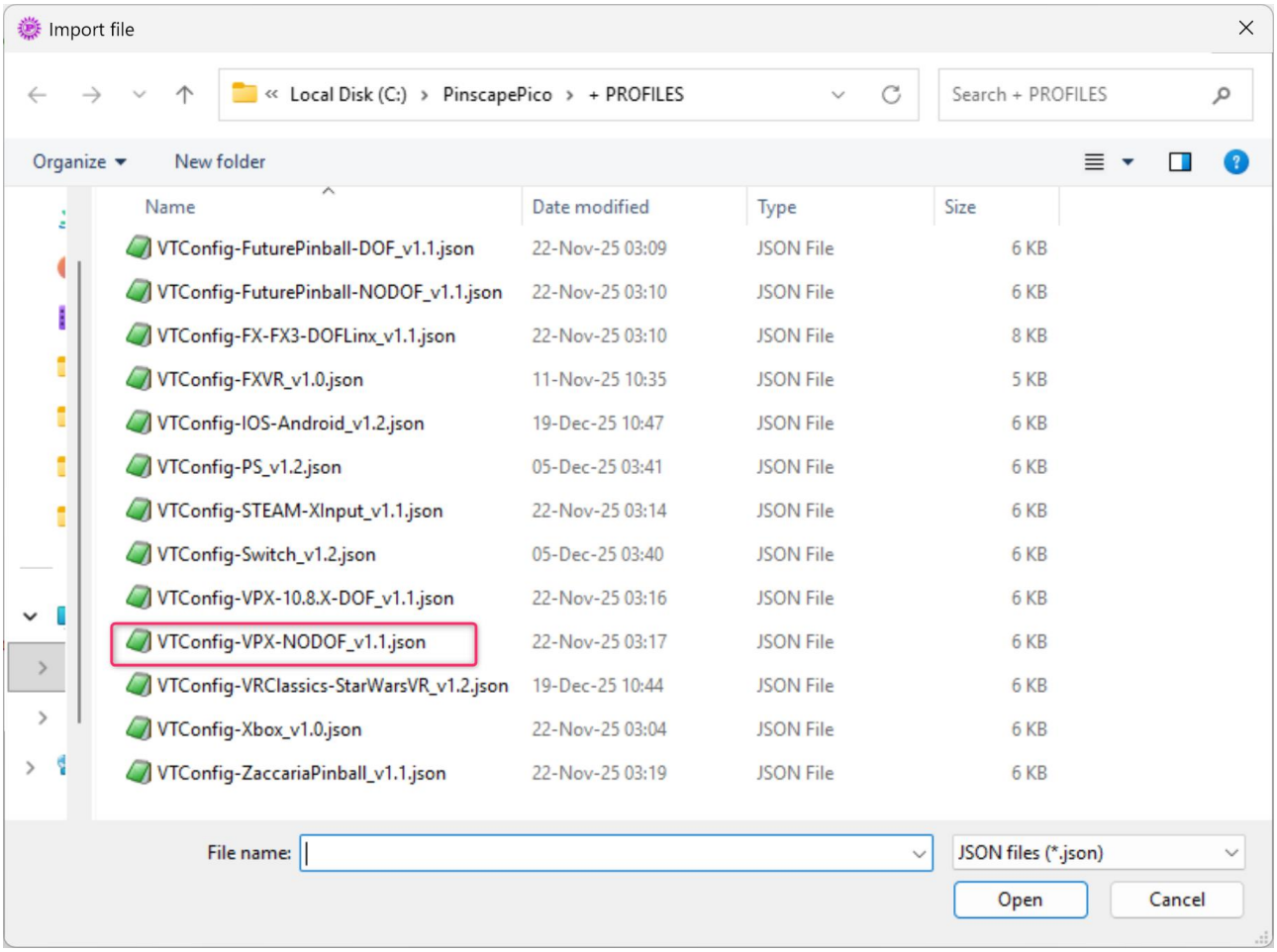


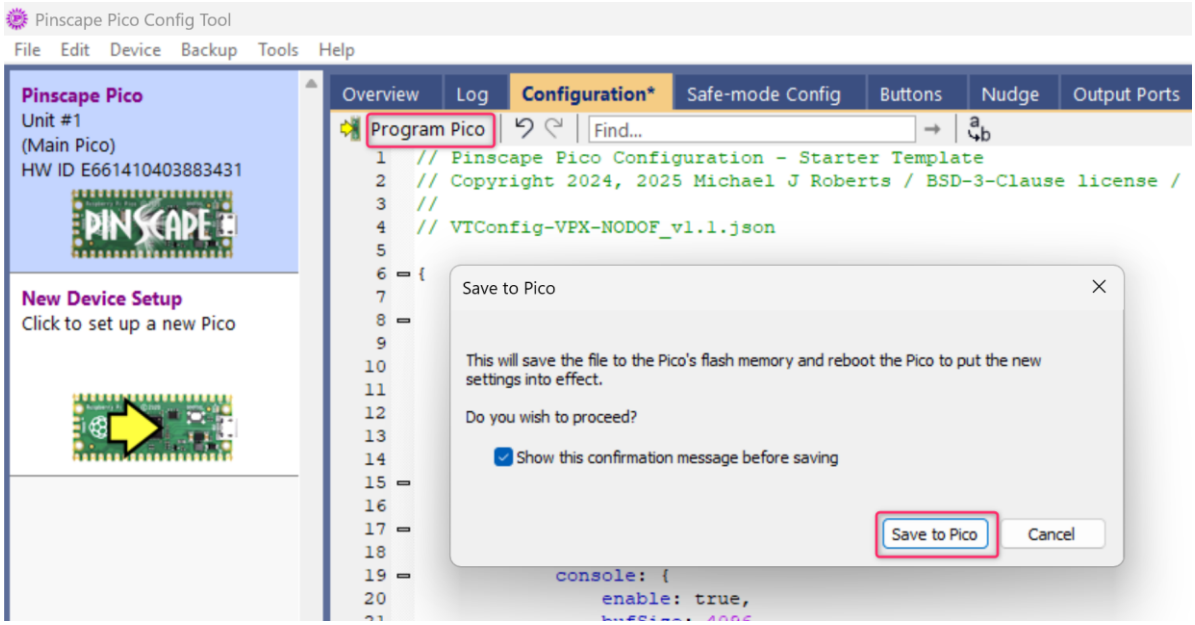
Play Visual Pinball X without DOF

The new upgraded VirtuaTilt allows you to enjoy feedback from flipper's solenoids and shaker motor even without DOF configured!

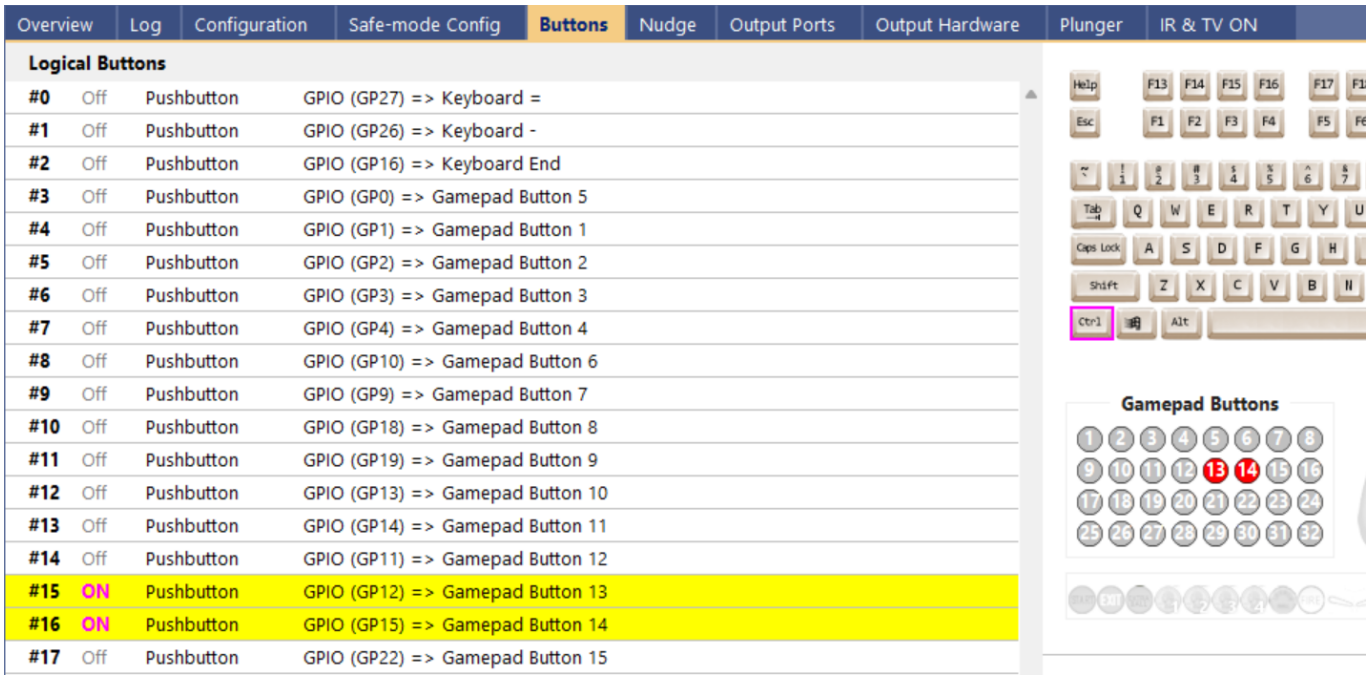
Load **VTConfig-VPX-NODOF_v1.X.json** in **Pinscape Pico Config Tool:**



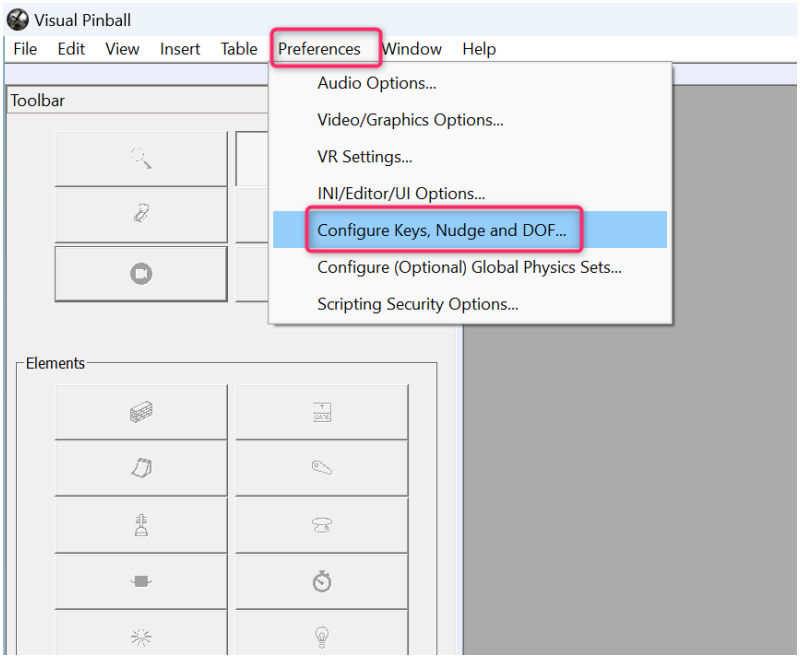


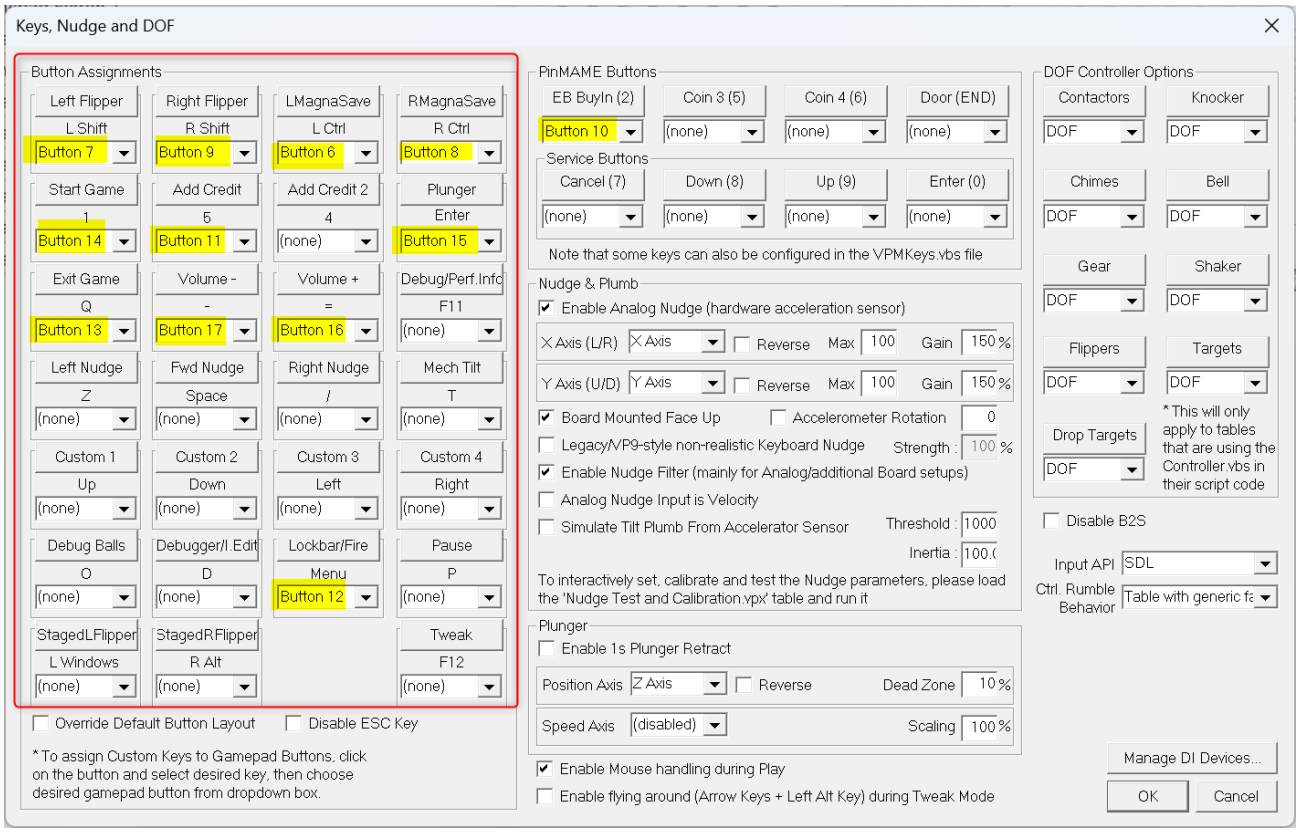


In Pinscape Pico config tool go to **Buttons** tab and note your buttons numbers (press every button)**:**

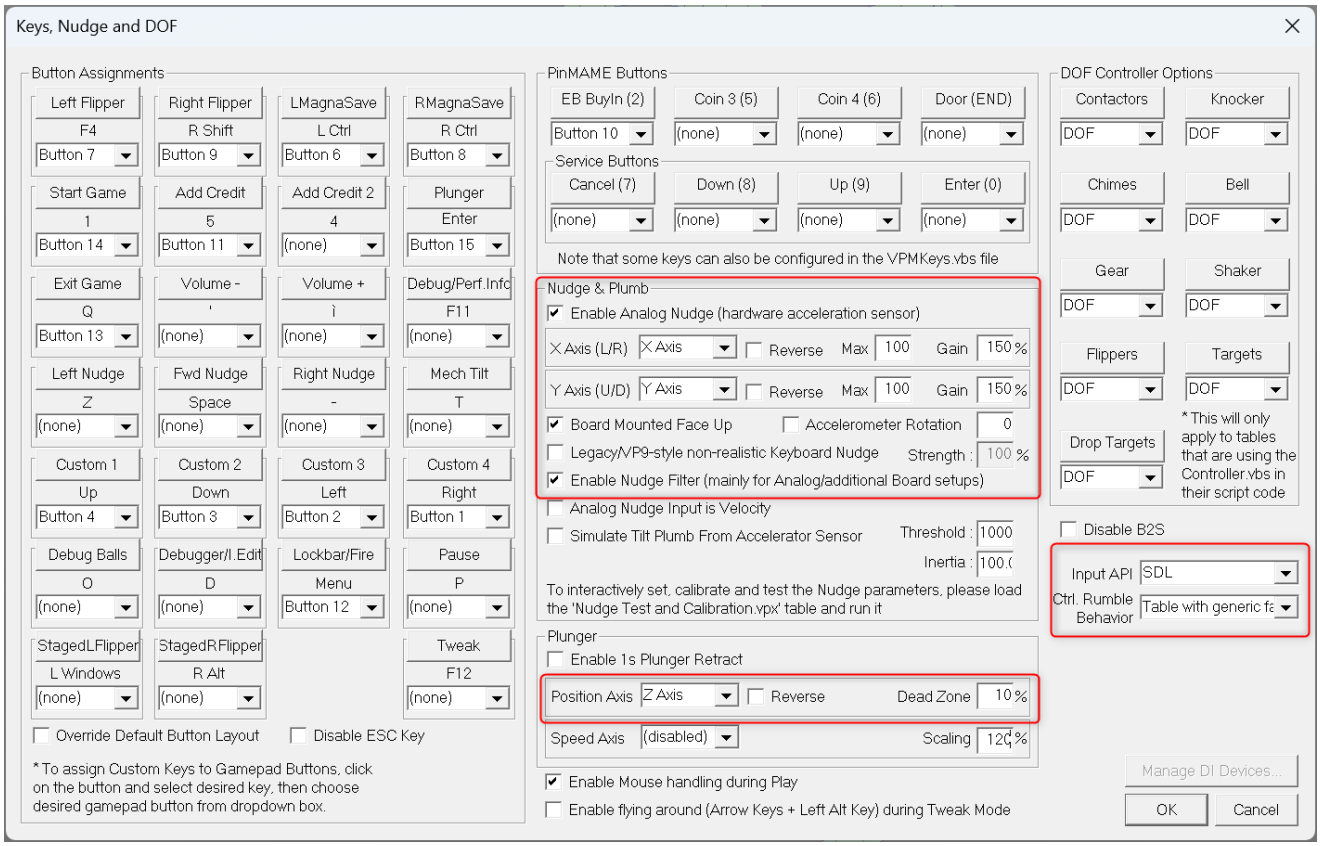


Assign all buttons you need in VPX:



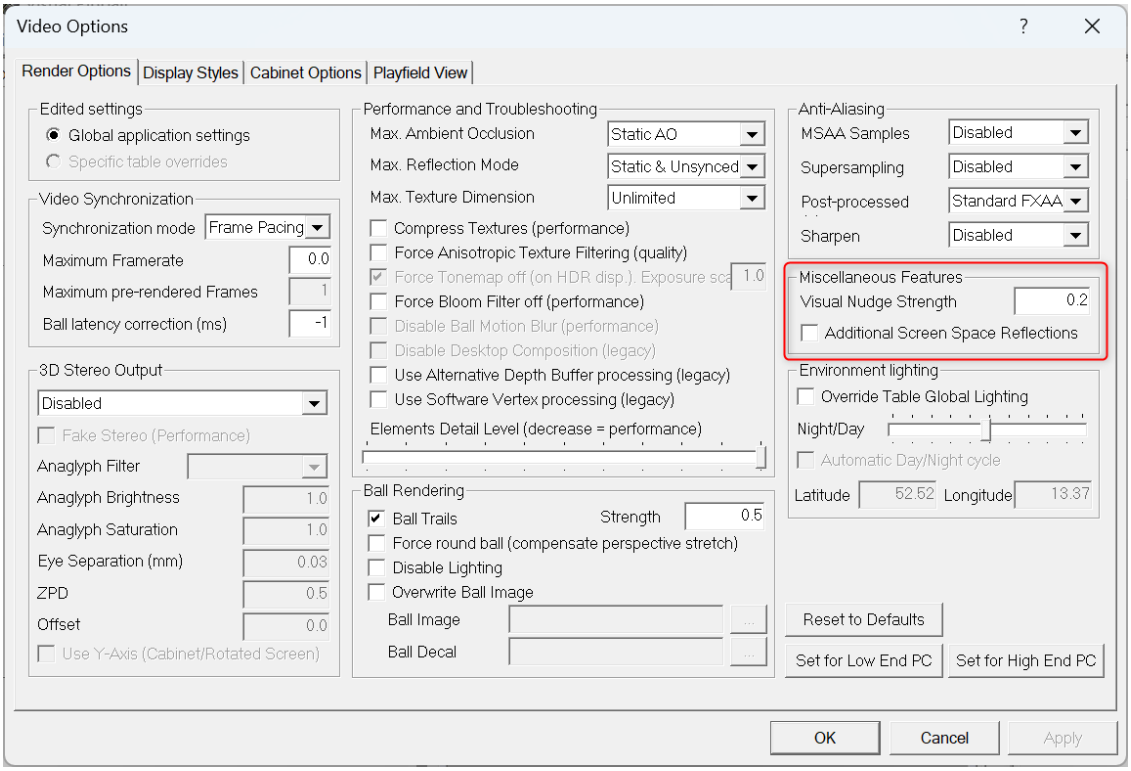


Replicate also the following configuration in Visual Pinball X settings (**VPX 10.8.1.XXXX**):



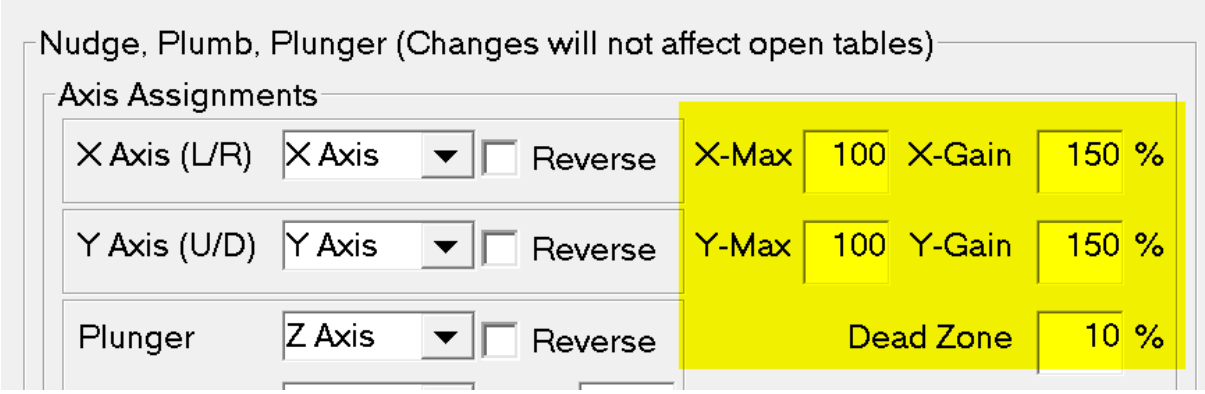
Nudge/Tilt configuration

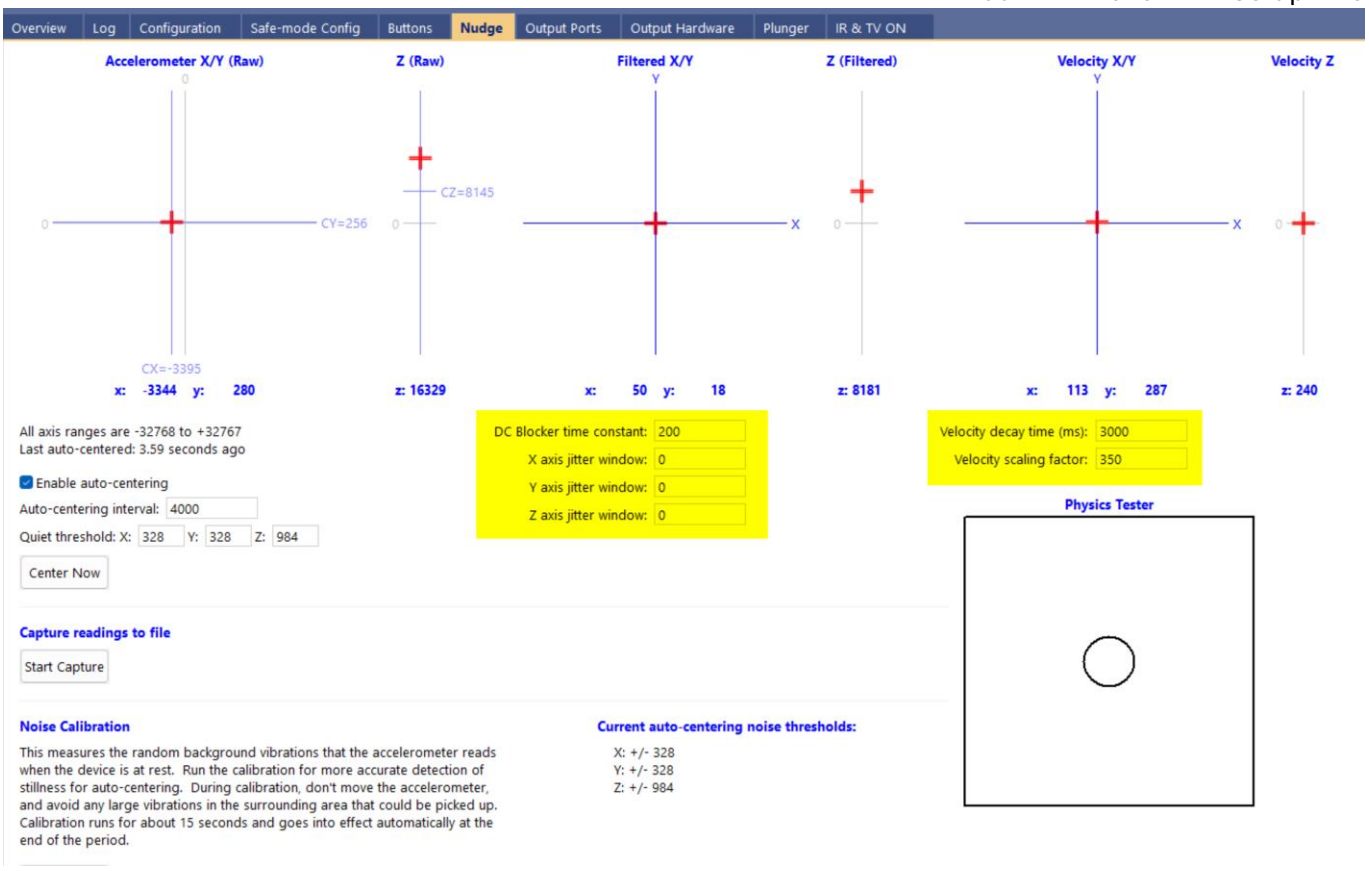
We recommend starting with the following values and modifying them afterwards:



Visual Nudge Strength: **0.2**

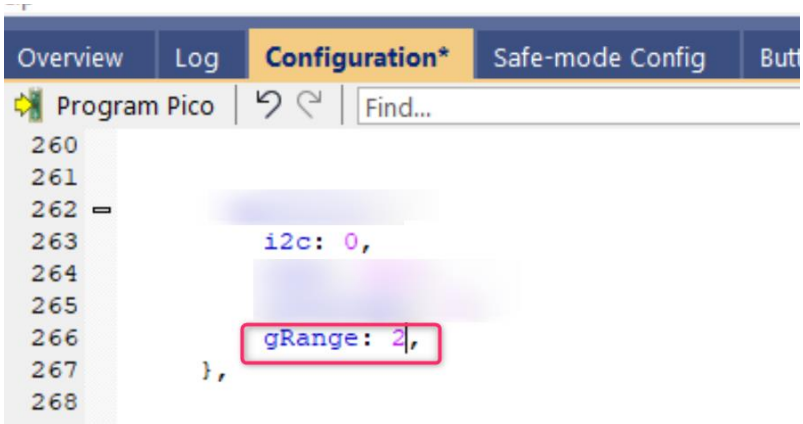
VERY IMPORTANT NOTE: You need to find your right settings (according to your personal preferences) for nudging. You can adjust both Pinscape Pico accelerometer settings and VPX settings until you find your perfect spot:





Try to adjust DC Blocker time constant, Velocity decay time and Velocity scaling factor.

You can also increase accelerometer sensitivity by changing gRange value to 2 in any profile:



Please refer to the official Pinscape Pico reference for additional information about accelerometer parameters: http://mjrnet.org/pinscape/PinscapePico/Help/NudgeWin.htm